

# An exact gamma constant for an Apéry-like sequence

A279619: two independent proofs, all-orders asymptotics,  
and a recessive companion

Research report prepared for Vladimir Reshetnikov

October 1, 2026

Merged from two independent manuscripts of batch 73O2 of ProveIt's incoming-reports intake (manuscripts 58 and 56; Section 15)

$$a_n \sim \frac{\sqrt{3}\pi}{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)} \frac{27^n}{n^{3/2}}.$$

The leading constant is evaluated exactly, the displayed OEIS asymptotic is proved, and the expansion is extended to arbitrary fixed order with a rigorous remainder. A rationally normalized companion exposes an exponentially smaller alternating scale.

**Abstract.** We study the level-seven Apéry-like sequence A279619, defined by a second-order polynomial recurrence. Its OEIS entry displays a conjectural asymptotic with a numerical constant. We prove that asymptotic and identify the constant as a reciprocal product of three gamma values. Two independent proofs of the constant are printed. The first: a rational parameter removes the square roots in the known hypergeometric generating function, making its singular connection coefficient explicit, and classical elliptic-period and Ramanujan identities reduce this coefficient to its gamma-product form. The second: the boundary value of the generating function is a Dedekind eta product at the Fricke fixed point  $i/\sqrt{7}$ , evaluated by the Chowla–Selberg formula at discriminant  $-7$ , and the singular coefficient is then read off from Hirschhorn's constant for the square sequence A183204. A functional equation gives a second exact sum, at  $z = 1/125$ . Frobenius expansion and singularity analysis give all algebraic correction terms, with rational coefficients and fixed-order error bounds, together with ratio, root and critical-tail expansions. We also construct a companion continued fraction, prove exact rational bracketing, and determine the amplitude and every correction term of its recessive solution. Applications include convolution powers, A183204, eventual log-convexity, and Lambert- $W$  inversion with a separate treatment of integer rounding. Exact symbolic certificates and rational enclosures accompany numerical experiments. Classical inputs, previously calculated terms, sequence-specific deductions, and unsettled questions are explicitly distinguished.

Shipped with this report: verification programs of both manuscripts, their recorded outputs, rational certificates, numerical tables, and manuscript 58's source audit (listed in the report's README.md).

These are conventional mathematical proofs with computer-algebra checks, not Lean-certified theorems. Global priority is not claimed.

## Contents

|          |                                                                              |           |
|----------|------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>The target and the scope of the contribution</b>                          | <b>4</b>  |
| 1.1      | What is already known . . . . .                                              | 4         |
| 1.2      | Relation to ProveIt . . . . .                                                | 5         |
| 1.3      | Two sources, one report . . . . .                                            | 5         |
| 1.4      | Notation of the two manuscripts . . . . .                                    | 5         |
| <b>2</b> | <b>Principal results</b>                                                     | <b>6</b>  |
| 2.1      | Immediate corollaries . . . . .                                              | 7         |
| <b>3</b> | <b>From the recurrence to a controlled singularity</b>                       | <b>9</b>  |
| 3.1      | Positivity, growth, and the differential equation . . . . .                  | 9         |
| 3.2      | The local exponents . . . . .                                                | 9         |
| <b>4</b> | <b>A rational certificate for the generating function</b>                    | <b>9</b>  |
| <b>5</b> | <b>Evaluating the connection constant (first proof)</b>                      | <b>10</b> |
| 5.1      | An explicit hypergeometric expression . . . . .                              | 10        |
| 5.2      | Ramanujan’s identity removes the derivative . . . . .                        | 11        |
| 5.3      | An elliptic period supplies the gamma product . . . . .                      | 12        |
| <b>6</b> | <b>Second proof: the square sequence and a CM evaluation (manuscript 56)</b> | <b>12</b> |
| 6.1      | The square identity . . . . .                                                | 13        |
| 6.2      | The modular parametrization . . . . .                                        | 13        |
| 6.3      | The square sequence by Laplace’s method . . . . .                            | 14        |
| 6.4      | The Fricke fixed point . . . . .                                             | 15        |
| 6.5      | Chowla–Selberg at discriminant $-7$ . . . . .                                | 15        |
| 6.6      | A second exact series from the functional equation . . . . .                 | 16        |
| 6.7      | The singular connection coefficient . . . . .                                | 16        |
| <b>7</b> | <b>All orders: existence, remainders, and coefficients</b>                   | <b>17</b> |
| 7.1      | The convergent local expansion . . . . .                                     | 17        |
| 7.2      | Transfer to coefficients . . . . .                                           | 18        |
| 7.3      | A triangular recurrence at infinity . . . . .                                | 18        |
| 7.4      | Logarithms, ratios, and convexity . . . . .                                  | 19        |
| <b>8</b> | <b>A continued fraction and exact rational bracketing</b>                    | <b>20</b> |
| 8.1      | The discrete Wronskian . . . . .                                             | 20        |
| 8.2      | Polynomial continued-fraction coefficients . . . . .                         | 20        |

|           |                                                                    |           |
|-----------|--------------------------------------------------------------------|-----------|
| <b>9</b>  | <b>The recessive scale and its full expansion</b>                  | <b>21</b> |
| 9.1       | A remainder proof for the geometric tail . . . . .                 | 21        |
| 9.2       | What the two-scale expansion does and does not say . . . . .       | 22        |
| <b>10</b> | <b>Convolution powers and A183204</b>                              | <b>22</b> |
| <b>11</b> | <b>Inversion and the integer-rounding boundary</b>                 | <b>23</b> |
| 11.1      | A triangular inverse expansion . . . . .                           | 23        |
| 11.2      | A rigorous discrete statement . . . . .                            | 24        |
| <b>12</b> | <b>Reproducible computations and certified constants</b>           | <b>25</b> |
| 12.1      | Certificates and reproducible output . . . . .                     | 25        |
| 12.2      | A rational enclosure for the asymptotic constant . . . . .         | 25        |
| 12.3      | Numerical diagnostics . . . . .                                    | 26        |
| 12.4      | Manuscript 56's tables and checks . . . . .                        | 27        |
| <b>13</b> | <b>Further research questions and topics</b>                       | <b>28</b> |
| 13.1      | Explicit remainder constants . . . . .                             | 28        |
| 13.2      | Late-order growth and optimal truncation . . . . .                 | 28        |
| 13.3      | Canonical exponentially improved asymptotics . . . . .             | 29        |
| 13.4      | Recognizing the continued-fraction limit . . . . .                 | 29        |
| 13.5      | Arithmetic strength of the approximants . . . . .                  | 29        |
| 13.6      | The arithmetic conjecture in the entry . . . . .                   | 29        |
| 13.7      | Coefficient fields for convolution powers . . . . .                | 29        |
| 13.8      | Noninteger powers and growing convolution order . . . . .          | 29        |
| 13.9      | Positive integral representations . . . . .                        | 30        |
| 13.10     | Automated connection-constant recognition . . . . .                | 30        |
| 13.11     | A direct modular proof of the product identity . . . . .           | 30        |
| 13.12     | Machine-checked certificates and analytic infrastructure . . . . . | 30        |
| 13.13     | Manuscript 56's questions . . . . .                                | 30        |
| <b>14</b> | <b>Scope, audits, and conclusion</b>                               | <b>32</b> |
| 14.1      | Manuscript 56's scope and novelty audit . . . . .                  | 32        |
| 14.2      | Integration roadmap and conclusion . . . . .                       | 32        |
| <b>15</b> | <b>Provenance of this report</b>                                   | <b>33</b> |
| <b>A</b>  | <b>Checking the classical period normalization</b>                 | <b>33</b> |
| <b>B</b>  | <b>Two derivations from manuscript 56</b>                          | <b>34</b> |

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| B.1      | The recurrence-to-ODE identity . . . . .                    | 34        |
| B.2      | A compact derivation of the coefficient algorithm . . . . . | 34        |
| <b>C</b> | <b>Possible OEIS formula updates</b>                        | <b>34</b> |
| C.1      | Manuscript 58's proposal . . . . .                          | 35        |
| C.2      | Manuscript 56's proposal . . . . .                          | 35        |

# 1 The target and the scope of the contribution

The sequence begins

$$1, 2, 22, 336, 6006, 117348, 2428272, 52303680, 1160427510, \dots$$

and satisfies

$$(n+1)^2 a_{n+1} = (26n^2 + 13n + 2)a_n + 3(3n-1)(3n-2)a_{n-1}, \quad n \geq 1, \quad (1)$$

with  $a_0 = 1, a_1 = 2$ . (Manuscript 56 writes the last coefficient as  $27n^2 - 27n + 6$ .) Throughout, set

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad \rho = 1/27, \quad \mathcal{G}_7 = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7). \quad (2)$$

This is A279619;  $A(z)^2$  is the generating function of A183204 [1, 2]. Its connections with modular forms, hypergeometric functions, and another Apéry-like sequence make it a structured target rather than an isolated numerical curiosity.

## 1.1 What is already known

The OEIS entry consulted on October 1, 2026 labels its leading asymptotic as conjectural and supplies a numerical constant. O’Brien’s thesis [3, §8.4, pp. 63–67] obtained the recurrence and the level-seven modular parametrization, estimates that constant and formally calculates the first three correction terms. Those three terms are *not new here*. The hypergeometric representation is also recorded in OEIS in code attributed to Václav Kotešovec in 2018 [1]. We prove and rationalize that representation before using it analytically. The leading asymptotic of the square sequence A183204, with a complete expansion, is Hirschhorn’s theorem [4]; it is used, and its leading constant re-derived, in the second proof (Section 6).

The contribution is not a new general method of asymptotic analysis. It is an exact connection-constant evaluation, a rigorous all-orders fixed-order expansion, and an explicit recessive companion with a proved error amplitude. Standard machinery is used to close the displayed asymptotic question.

| Item                                      | Status in this report                                                                                                                                     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recurrence and hypergeometric function    | Existing input; the latter receives a rational differential-equation certificate.                                                                         |
| Leading scale and first three corrections | Previously conjectured or formally calculated; now given a remainder proof.                                                                               |
| Gamma-product constant                    | Exact sequence-specific evaluation deduced from classical special values, by two independent routes.                                                      |
| Sum at $z = 1/125$                        | Deduced, in manuscript 56, from the constant and a functional equation.                                                                                   |
| Companion continued fraction              | Constructed with exact bracketing and all-orders recessive asymptotics; global priority not asserted.                                                     |
| Convolution and inverse expansions        | Deductions with explicit formulas and integer-rounding bounds; the inverse expansion is an instance of the repository’s inversion apparatus (Section 11). |
| Lucas congruences                         | Not resolved or used here.                                                                                                                                |

A targeted search did not locate an earlier proof of this particular constant formula. This is not an exhaustive priority search. In particular, recent work of Cohen and Zudilin on continued fractions for Chowla–Selberg quotients [14] shows that the broader area is active. We do not claim a first continued fraction for a gamma quotient, an identification of the companion limit with such a quotient, or an irrationality measure. Manuscript 56’s own, independent audit is printed in Section 14.1.

## 1.2 Relation to ProveIt

The repository’s Analysis/Transseries project separates asymptotic scales, flatness, power–logarithmic expressions, and inversion from the Fabius-function development [15]. This report follows that separation: an analytic existence proof precedes exact coefficient recursions and inverse expansions. No unverified repository theorem is needed. The inspected snapshot was a5ad4ac2e2d4e642a6c015836a70492d40fcb71; inspection was not an exhaustive review of archived reports.

[Merge note, batch 7302: The Lambert- $W$  inverse of Section 11 is the instance  $a = \log 27$ ,  $\beta = -3/2$  of Theorem t2:thm:balanced-inverse of the repository’s volume *Combinatorial Transseries and Their Inverses* [16], which neither manuscript cites; it is printed as an application, with no novelty claimed for it. Manuscript 58’s rounding proposition is the sequence-specific part.]

## 1.3 Two sources, one report

This report merges two manuscripts that reached ProveIt on the same day, October 1, 2026, in batch 7302 of its incoming-reports intake, and that were written independently: manuscript 58 (*An exact gamma constant for an Apéry-like sequence*) and manuscript 56 (*Exact asymptotics of OEIS A279619*). Manuscript 58 inspected the repository at a commit made before manuscript 56 arrived, and manuscript 56 cites neither manuscript 58 nor any repository revision. Both prove the same leading constant, the same boundary value  $A(\rho)$ , and the same rational corrections (they agree as rationals through order 10, the last order manuscript 56 prints; manuscript 58 records them to order 12). They do so by *different* proofs. Two independent derivations of the same constant arriving on the same day is mild corroboration, not a substitute for review.

Manuscript 58 is the base text: Sections 1–5 and 7–12 are its text, with its labels prefixed, its hard-coded section numbers replaced by references, and manuscript 56’s additions inserted as marked passages. Manuscript 56 contributes the second proof of the constant (Section 6), the sum at  $1/125$  and the critical-tail, ratio and root corollaries (Section 2), the integer form of the Frobenius recurrence and the coefficients  $c_6, \dots, c_{10}$  (Section 7), its inverse remark (Section 11), its numerical tables (Section 12), ten research problems (Section 13.13), its novelty audit (Section 14.1), two derivations (Appendix B) and its OEIS proposal (Appendix C). Results proved in both manuscripts are printed once and marked “proved in both sources”. Section 15 records the provenance.

## 1.4 Notation of the two manuscripts

The two manuscripts use several letters for different objects. This report uses manuscript 58’s notation throughout; manuscript 56’s material has been rewritten in it. The dictionary, with the tempting false readings, is Table 1.

Table 1: Notation of manuscripts 58 and 56.

| Object                      | This report (= 58)                | Manuscript 56         | Caution                                                     |
|-----------------------------|-----------------------------------|-----------------------|-------------------------------------------------------------|
| the sequence A279619        | $a_n$                             | $c_n$                 | here $c_j$ are <i>corrections</i>                           |
| its generating function     | $A(z)$                            | $F(x)$                | here $F$ is ${}_2F_1(\frac{1}{12}, \frac{5}{12}; 1; \cdot)$ |
| gamma product               | $\mathcal{G}_7$                   | $\mathcal{G}_7$       | same                                                        |
| leading constant            | $C$                               | $C_c$                 | —                                                           |
| boundary value $A(\rho)$    | $A_\star$                         | $F(1/27), A(0)$       | 56's $A(s)$ is not the GF                                   |
| corrections                 | $c_j$                             | $\alpha_j$            | equal as rationals                                          |
| analytic part at $\rho$     | $U(s)$                            | $A(s)$                | —                                                           |
| singular coefficient        | $B$                               | $B_0$                 | —                                                           |
| singular Frobenius series   | $V(s) = \sum v_j s^j$             | $B(s) = \sum b_m s^m$ | here $b_n$ is the <i>companion</i>                          |
| square sequence (A183204)   | $t_n = a_n^{(2)}, A(z)^2$         | $t_n, G(x) = F(x)^2$  | $G$ is not $\mathcal{G}_7$                                  |
| its leading constant        | $C_t = 2CA_\star$                 | $C_t$                 | —                                                           |
| log 27                      | $\lambda$                         | $L$                   | here $L$ is the companion limit                             |
| entropy function            | $\text{Ent}(u)$                   | $H(u)$                | here $H$ is Clausen's ${}_3F_2$                             |
| Laplace variable, weight    | $\xi = k/n, \kappa(\xi)$          | $a, g(a)$             | $a$ is not a term                                           |
| scaled term                 | $R_n = a_n n^{3/2}/27^n$          | $Q_n$                 | here $Q_n, Q(m)$ differ                                     |
| critical partial sum, shift | $\Sigma_N, \nu = N + 1$           | $S_N, q$              | here $S_N$ sums $F(q_0)$                                    |
| continuous inverse          | $x_0(Y)$                          | $n_0(y)$              | —                                                           |
| formal series at infinity   | $S_{27}(t)$                       | $\mathcal{A}(t)$      | —                                                           |
| Weber invariant power       | $u = \mathfrak{f}_2(\tau_0)^{24}$ | $u$                   | here $u_n$ is a solution                                    |
| level-7 Hauptmodul          | $w_7(\tau)$                       | $w_7(\tau)$           | here $w$ is the rational parameter                          |
| modular $j$ -invariant      | $j(\tau)$                         | $j(\tau)$             | here $j(m)$ is a pullback                                   |

No normalization changes:  $C = C_c$ ,  $B = B_0$  (both with the singular Frobenius series normalized to 1 at  $s = 0$ ) and  $\lambda = L_{56}$ . Manuscript 56's index range “ $M \geq 1$ , error  $O(n^{-M})$ , sum to  $M - 1$ ” is manuscript 58's “ $M \geq 0$ , error  $O(n^{-M-1})$ , sum to  $M$ ” shifted by one.

## 2 Principal results

**Theorem 2.1** (Dominant asymptotics and the exact constant; proved in both sources). *Put*

$$C = \frac{\sqrt{3\pi}}{\mathcal{G}_7}, \quad A_\star = \frac{3\mathcal{G}_7}{8\pi^2}. \quad (3)$$

Then  $A(\rho) = A_\star$ . For every fixed  $M \geq 0$ ,

$$a_n = C \frac{27^n}{n^{3/2}} \left( \sum_{j=0}^M \frac{c_j}{n^j} + O_M(n^{-M-1}) \right), \quad (4)$$

where the  $c_j$  are uniquely determined rational numbers with  $c_0 = 1$ . In particular,

$$\begin{aligned} c_1 &= -\frac{215}{1008}, & c_2 &= -\frac{1265}{290304}, & c_3 &= \frac{4683055}{877879296}, \\ c_4 &= \frac{924908071}{3539609321472}, & c_5 &= -\frac{6088275239005}{3567926196043776}. \end{aligned} \quad (5)$$

An exact triangular algorithm computes every coefficient.

The theorem also evaluates a convergent boundary sum:

$$\sum_{n=0}^{\infty} \frac{a_n}{27^n} = \frac{3\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)}{8\pi^2}, \quad CA_{\star} = \frac{3\sqrt{3}}{8\pi^{3/2}}. \quad (6)$$

The gamma product in the boundary value cancels against its reciprocal in the coefficient amplitude. Numerically  $C = 0.0955223052681267146513079107870296256727946666510071798669 \dots$  and  $A_{\star} = 1.221131097467566669284224585685993462375827056379 \dots$ ; the two manuscripts' decimal values agree in every printed digit.

[Merge note, batch 7302: Manuscript 58 proves Theorem 2.1 in Sections 3–7; manuscript 56 proves it in Section 6 and in its part of Section 7. Manuscript 56's statement is the same theorem with  $\alpha_j = c_j$  and the shifted index range of Section 1.4; it lists the same five coefficients.]

**Theorem 2.2** (Exact critical and subcritical sums; manuscript 56). *The generating function has the exact values*

$$A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}, \quad A(1/125) = \sum_{n=0}^{\infty} \frac{a_n}{125^n} = \frac{5\mathcal{G}_7}{16\pi^2}. \quad (7)$$

Numerically,  $A(1/125) = 1.017609247889638891070187154738327885313189213649 \dots$

The first value is (6). The second is proved in Section 6.6 from the first and a functional equation; it appears only in manuscript 56.

**What is resolved** (manuscript 56). The decimal constant in the OEIS asymptotic conjecture is exactly the gamma quotient in (3). More is true: the complete expansion exists to arbitrary algebraic order, all its normalized coefficients are rational, the singular connection coefficient is explicit, and the critical value of the generating function is a closed gamma product.

## 2.1 Immediate corollaries

The all-orders form gives several useful consequences without further modular input. The three corollaries below are manuscript 56's.

**Corollary 2.3** (Ratio asymptotics and eventual log-convexity). *As  $n \rightarrow \infty$ ,*

$$\frac{a_{n+1}}{a_n} = 27 \left( 1 - \frac{3}{2n} + \frac{2105}{1008n^2} - \frac{18815}{7056n^3} + \frac{3425255}{1053696n^4} + O(n^{-5}) \right). \quad (8)$$

*Consequently  $a_{n+1}/a_n \rightarrow 27$ , and  $(a_n)$  is strictly log-convex for all sufficiently large  $n$ .*

*Proof.* Insert (4) into the quotient. The difference of successive ratios is

$$\frac{a_{n+2}}{a_{n+1}} - \frac{a_{n+1}}{a_n} = \frac{81}{2n^2} + O(n^{-3}) > 0$$

for all sufficiently large  $n$ . Increasing ratios are equivalent to strict log-convexity. □

Manuscript 58 proves the first three terms of (8) and eventual log-convexity through second differences of  $\log a_n$  (Section 7.4); the two increments are consistent,  $81/(2n^2) \div 27 = 3/(2n^2)$ .

**Corollary 2.4** (Root and logarithmic scales). *With  $\lambda = \log 27$ ,*

$$\log a_n = n\lambda - \frac{3}{2} \log n + \log C - \frac{215}{1008n} + O(n^{-2}), \quad (9)$$

$$a_n^{1/n} = 27 \exp\left(\frac{\log C - \frac{3}{2} \log n}{n} + O(n^{-2})\right). \quad (10)$$

The first line is the case  $M = 1$  of manuscript 58's logarithmic expansion (77).

**Corollary 2.5** (Critical-tail asymptotics). *Let  $\Sigma_N = \sum_{n=0}^N a_n/27^n$  and  $\nu = N + 1$ . Then*

$$A(1/27) - \Sigma_N = C \left( 2\nu^{-1/2} + \frac{541}{1512} \nu^{-3/2} + \frac{2411}{145152} \nu^{-5/2} + O(\nu^{-7/2}) \right). \quad (11)$$

*In particular, convergence at the critical point is only of order  $N^{-1/2}$ .*

*Proof.* Sum (4) termwise and use the standard Hurwitz-zeta expansion for  $\zeta(s, N + 1)$ . □

[Merge note, batch 73O2: The intake re-derived 541/1512 and 2411/145152 by the Euler–Maclaurin formula, and checked that the partial sum through  $N = 5000$  plus the three-term tail matches  $3\mathcal{G}_7/(8\pi^2)$  to  $4.9 \times 10^{-16}$ , consistent with the stated  $O(\nu^{-7/2})$ .]

**Theorem 2.6** (An exact recessive companion; manuscript 58). *Let  $b_0 = 0$ ,  $b_1 = 1$ , and impose (1) on  $b_n$  for  $n \geq 1$ . Then the limit*

$$L = \lim_{n \rightarrow \infty} \frac{b_n}{a_n} = 0.46721176888437312394211005141309126172410854995971 \dots \quad (12)$$

*exists and satisfies*

$$\frac{b_{2m}}{a_{2m}} < L < \frac{b_{2m+1}}{a_{2m+1}}, \quad m \geq 0. \quad (13)$$

*For  $e_n = b_n - La_n$  and every fixed  $M \geq 0$ ,*

$$e_n = (-1)^{n+1} \frac{\sqrt{3}}{56\pi C} n^{-3/2} \left( \sum_{j=0}^M \frac{d_j}{n^j} + O_M(n^{-M-1}) \right), \quad (14)$$

*where  $d_0 = 1$  and*

$$d_1 = -\frac{57}{112}, \quad d_2 = \frac{655}{3584}, \quad d_3 = -\frac{13365}{401408}, \quad d_4 = -\frac{3627123}{179830784}. \quad (15)$$

*The  $d_j$  are rational and computable to arbitrary order. Consequently,*

$$L - \frac{b_n}{a_n} \sim (-1)^n \frac{\sqrt{3}}{56\pi C^2} 27^{-n}. \quad (16)$$

The two scales are  $27^n n^{-3/2}$  and  $(-1)^n n^{-3/2}$ . The second is exponentially small *relative* to the first, not in absolute magnitude. We do not assign an unproved canonical subdominant coefficient to the asymptotic expansion of  $a_n$  itself.

### 3 From the recurrence to a controlled singularity

#### 3.1 Positivity, growth, and the differential equation

The recurrence immediately gives  $a_n > 0$ . Induction also gives  $a_n \leq 27^n$ , since

$$26n^2 + 13n + 2 + \frac{3(3n-1)(3n-2)}{27} = 27n^2 + 12n + \frac{20}{9} < 27(n+1)^2.$$

Thus  $A$  is analytic in  $|z| < \rho$ . Multiplying the recurrence by  $z^n$  and summing, including  $a_1 = 2a_0$ , gives

$$zD(z)A'' + (1 - 39z - 54z^2)A' - (2 + 6z)A = 0, \quad D(z) = (1 - 27z)(1 + z). \quad (17)$$

The coefficient of  $z^n$  on the left is the recurrence residual, so the analytic solution normalized by  $A(0) = 1$  is unique. Outside  $0, \rho, -1$ , the equation is ordinary; analytic continuation along paths avoiding these points follows from linear differential-equation theory. (Manuscript 56 derives the same equation, written with  $zD(z) = z(1 - 26z - 27z^2)$ , in Appendix B. Its positive singularity  $\rho$  is separated from the other nonzero finite singularity  $-1$  by a factor of 27 in modulus.)

#### 3.2 The local exponents

In  $s = 1 - 27z$ , the equation becomes

$$s(1-s)(28-s)y'' + (14 - 43s + 2s^2)y' - \frac{20-2s}{9}y = 0. \quad (18)$$

The indicial equation is  $14\alpha(2\alpha - 1) = 0$ . Its exponents  $0, 1/2$  have nonintegral difference, hence Frobenius theory gives a convergent local expansion

$$A(z) = U(s) + B\sqrt{s}V(s), \quad V(0) = 1, \quad (19)$$

where  $U, V$  are analytic at 0. Branches are positive for  $0 < z < \rho$ . Below we show  $B < 0$ , so the singularity is not removable.

The self-adjoint form provides a useful sign check:

$$(z\sqrt{D(z)}A'(z))' = \frac{2+6z}{\sqrt{D(z)}}A(z), \quad 0 < z < \rho. \quad (20)$$

Frobenius theory makes  $A(\rho)$  finite. Positivity of the right side implies that the limiting flux is positive, whereas (19) identifies it as  $-\frac{1}{2}\sqrt{28/27}B$ .

There is no other singularity on  $|z| = \rho$ . The chosen solution extends to a dented disk of radius greater than  $\rho$ , cut to the right from  $\rho$ . This supplies the global analytic condition for singularity analysis, not just a formal local expansion [12, Chapter VI].

### 4 A rational certificate for the generating function

Set

$$w = \sqrt{\frac{1-27z}{1+z}}, \quad z = \frac{1-w^2}{27+w^2}. \quad (21)$$

The points  $0, \rho$  correspond to  $w = 1, 0$ . Define

$$\begin{aligned} T(w) &= 15w^2 + 96w + 85, \\ P(w) &= \left( \frac{7(w^2 + 27)}{T(w)} \right)^{1/4}, \\ J(w) &= \frac{1728(1-w)^7(1+w)}{(w^2 + 27)T(w)^3}, \quad F(x) = {}_2F_1 \left( \begin{matrix} 1/12, 5/12 \\ 1 \end{matrix}; x \right). \end{aligned} \quad (22)$$

Fourth roots are positive on  $[0, 1]$ .

**Proposition 4.1.** *As analytic germs near  $w = 1$ , and throughout  $0 \leq w \leq 1$ ,*

$$A \left( \frac{1-w^2}{27+w^2} \right) = P(w)F(J(w)). \quad (23)$$

*Proof.* The equation for  $y(w) = A(z(w))$  is  $y'' + py' + qy = 0$ , where

$$p = \frac{2w}{(w-1)(w+1)}, \quad q = -\frac{16(w^2-15)}{(w-1)(w+1)(w^2+27)^2}. \quad (24)$$

Let  $h = P'/P = \frac{1}{4}(2w/(w^2+27) - T'/T)$ . The hypergeometric equation [13, §15.10] is

$$x(1-x)F'' + (1-\frac{3}{2}x)F' - \frac{5}{144}F = 0. \quad (25)$$

Substitution of  $y = PF(J)$ , followed by eliminating  $F''$ , reduces the claim to two rational identities:

$$\begin{aligned} J'' + (2h+p)J' + (J')^2 \frac{3J/2-1}{J(1-J)} &= 0, \\ h' + h^2 + ph + q + \frac{5}{144} \frac{(J')^2}{J(1-J)} &= 0. \end{aligned} \quad (26)$$

Clearing denominators and expanding proves both; the attached code checks them identically in  $\mathbb{Q}(w)$ . This is an exact certificate rather than a fit to finitely many coefficients. At  $w = 1$ ,  $P = 1$ ,  $J = 0$ , and  $z'(1) \neq 0$ . The right side is analytic there and normalized to 1. Uniqueness at  $z = 0$  proves the germ identity. For  $0 \leq w < 1$ ,

$$\frac{J'}{J} = -\frac{7}{1-w} + \frac{1}{1+w} - \frac{2w}{w^2+27} - 3\frac{30w+96}{T(w)} < 0.$$

Thus  $0 \leq J(w) \leq J(0) = 64/85^3 < 1$ , and every expression continues regularly along the interval, proving the full claim.  $\square$

Differentiation at a singular  $z$ -value has now become ordinary differentiation at the regular parameter value  $w = 0$ .

## 5 Evaluating the connection constant (first proof)

### 5.1 An explicit hypergeometric expression

Put

$$q_0 = \frac{64}{85^3}, \quad P_0 = \left( \frac{189}{85} \right)^{1/4}. \quad (27)$$

As  $s \rightarrow 0$ ,

$$w = \sqrt{\frac{27s}{28-s}} = \sqrt{\frac{27}{28}}\sqrt{s}(1 + O(s)).$$

At  $w = 0$  we have  $P'/P_0 = -24/85$  and  $J'/q_0 = -798/85$ . Hence

$$B = -\sqrt{\frac{27}{28}} \frac{P_0}{85} (24F(q_0) + 798q_0F'(q_0)) < 0. \quad (28)$$

Positivity follows from the positive hypergeometric coefficients. Coefficient transfer for  $\sqrt{1-27z}$  and  $\Gamma(-1/2) = -2\sqrt{\pi}$  gives

$$C = -\frac{B}{2\sqrt{\pi}} = \sqrt{\frac{27}{28}} \frac{P_0}{170\sqrt{\pi}} (24F(q_0) + 798q_0F'(q_0)). \quad (29)$$

This already gives an explicit positive constant. The following special-value reductions explain its unexpectedly simple form.

## 5.2 Ramanujan's identity removes the derivative

The relevant Clausen identity is

$$F(q)^2 = H(q) := \sum_{n \geq 0} \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} q^n. \quad (30)$$

It follows directly from differential equations: the symmetric square of (25) is

$$[\vartheta^3 - q(\vartheta + 1/6)(\vartheta + 1/2)(\vartheta + 5/6)]H = 0, \quad \vartheta = q \frac{d}{dq},$$

whose analytic solution with  $H(0) = 1$  is unique. The rational coefficients of this symmetric-square identity are checked in the code.

The classical level-seven Ramanujan identity, reproduced and obtained by the construction in [6, Eq. (98)], is

$$\sum_{n \geq 0} (133n + 8) \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} q_0^n = \frac{85\sqrt{255}}{54\pi}. \quad (31)$$

The source uses the coefficient  $(6n)!/((3n)!(n!)^3)$ , which is  $1728^n$  times the coefficient here, and  $1728/255^3 = 64/85^3$ . This normalization check is essential. Equation (31) is a cited established identity, not an equality inferred from numerical recognition. Using (30), it becomes

$$8F(q_0)^2 + 266q_0F(q_0)F'(q_0) = \frac{85\sqrt{255}}{54\pi}.$$

Multiply by  $3/F(q_0)$  and substitute into (29). Since  $A_\star = P_0F(q_0)$  by Proposition 4.1, this yields

$$C = \frac{3\sqrt{3}}{8\pi^{3/2}P_0F(q_0)} = \frac{3\sqrt{3}}{8\pi^{3/2}A_\star}. \quad (32)$$

### 5.3 An elliptic period supplies the gamma product

Let

$$m_7 = \frac{8 - 3\sqrt{7}}{16}, \quad k_7 = \sqrt{m_7}, \quad K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}. \quad (33)$$

Here  $K$  takes the *modulus*  $k$ , not the parameter  $m = k^2$ . The classical singular-period evaluation is

$$K(k_7) = \frac{\mathcal{G}_7}{4\pi 7^{1/4}}. \quad (34)$$

This is an established complex-multiplication special value [7, 8], not a new identity claimed here. Appendix A explicitly reduces the beta-function form in Zucker’s table to this normalization.

The required hypergeometric transformation is

$$F\left(\frac{27m^2(1-m)^2}{4(1-m+m^2)^3}\right) = (1-m+m^2)^{1/4} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; m\right). \quad (35)$$

For a direct verification put  $Q(m) = 1 - m + m^2$  and  $j(m) = 27m^2(1-m)^2/(4Q(m)^3)$ . Then  $Q^{-1/4}F(j)$  satisfies

$$m(1-m)y'' + (1-2m)y' - \frac{1}{4}y = 0.$$

Substituting and clearing denominators proves two rational identities, exactly as in (26). Both solutions are analytic at 0 and equal to 1, proving the transformation near 0. Continuation on  $[0, m_7]$  proves the value used here. The program verifies both identities.

At  $m = m_7$ ,

$$m_7(1-m_7) = 1/256, \quad Q(m_7) = 255/256, \quad j(m_7) = q_0.$$

Euler’s integral gives  ${}_2F_1(1/2, 1/2; 1; m_7) = 2K(k_7)/\pi$ . Therefore

$$\begin{aligned} A_\star &= P_0 F(q_0) = \left(\frac{189}{85} \frac{255}{256}\right)^{1/4} \frac{2K(k_7)}{\pi} \\ &= \frac{3 \cdot 7^{1/4}}{4} \frac{2}{\pi} \frac{\mathcal{G}_7}{4\pi 7^{1/4}} = \frac{3\mathcal{G}_7}{8\pi^2}. \end{aligned}$$

Combining this with (32) proves  $C = \sqrt{3\pi}/\mathcal{G}_7$ . Since the coefficients are positive, the finite limit at  $\rho$  also gives the boundary sum (6), by monotone convergence.

*Remark 5.1* (Dependencies of the exact evaluation). The two non-elementary special values are (31) and (34); both are classical inputs. The recurrence manipulations, rational pullbacks, connection-coefficient extraction, and asymptotic deductions are proved here. We do not purport to give a new derivation of Chowla–Selberg theory or of Ramanujan’s general  $1/\pi$  formulas.

## 6 Second proof: the square sequence and a CM evaluation (manuscript 56)

This section is manuscript 56’s proof of  $A_\star = 3\mathcal{G}_7/(8\pi^2)$  and  $C = \sqrt{3\pi}/\mathcal{G}_7$ , rewritten in this report’s notation (Section 1.4). It is independent of Sections 4 and 5: it uses neither the  ${}_2F_1$  pullback, nor Ramanujan’s series, nor the elliptic period. Its classical inputs are O’Brien’s modular parametrization, the Chowla–Selberg formula at discriminant  $-7$ , and Hirschhorn’s asymptotic for A183204, whose leading constant it re-derives.

The proof combines four structures already present around this level-seven sequence: the square identity  $A(z)^2 = \sum t_n z^n$  with the Apéry-like sequence A183204, a one-dimensional Laplace analysis of the binomial sum for  $t_n$ , O'Brien's modular parametrization, and a Chowla–Selberg evaluation of a Dedekind eta value of discriminant  $-7$ . They come from two apparently different descriptions of the same object: a square root of an Apéry-like generating function and a modular parametrization on  $X_0(7)$ .

## 6.1 The square identity

Let

$$A(z)^2 = \sum_{n \geq 0} t_n z^n, \quad t_n = a_n^{(2)}. \quad (36)$$

O'Brien's level-seven construction, equivalently the relation recorded in OEIS, identifies  $(t_n)$  with A183204, one of the sporadic Apéry-like sequences [9]. (Manuscript 58 checks the identification through the symmetric-square equation (93).) Its coefficients have the finite binomial sum

$$t_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{2k}{n} \binom{n+k}{k}, \quad (37)$$

where terms with  $2k < n$  vanish. The initial values are

$$1, 4, 48, 760, 13840, 273504, \dots$$

[Merge note, batch 7302: Manuscript 58 checks the convolution against the different binomial form  $\sum_k \binom{n}{k}^2 \binom{n+k}{n} \binom{2n-k}{n}$ ; the intake checked that both forms equal  $[z^n]A(z)^2$  for  $n \leq 40$ .] Hirschhorn proved a complete asymptotic expansion for this sequence [4]. We rederive its leading constant in Section 6.3, because that constant is the bridge from the known square to the previously unknown square root.

## 6.2 The modular parametrization

Let  $\eta$  denote Dedekind's eta function [13, Ch. 23],

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{m \geq 1} (1 - e^{2\pi i m \tau}), \quad \Im \tau > 0. \quad (38)$$

O'Brien's parametrization may be written

$$w_7(\tau) = \left( \frac{\eta(7\tau)}{\eta(\tau)} \right)^4, \quad (39)$$

$$X_7(\tau) = \frac{w_7(\tau)}{1 + 13w_7(\tau) + 49w_7(\tau)^2}, \quad (40)$$

$$X_7(\tau) = \left( \frac{\eta(\tau)\eta(7\tau)}{z_7(\tau)} \right)^3, \quad A(X_7(\tau)) = z_7(\tau). \quad (41)$$

The last identity is initially a  $q$ -series identity near the cusp and then extends by analytic continuation. It is the crucial reason the critical value of  $A$  can be reduced to a CM eta product.

### 6.3 The square sequence by Laplace's method

We now obtain the leading asymptotic of  $t_n$  directly from (37). This gives a self-contained derivation of the one published constant from which the new constant will be extracted.

For  $\xi \in (1/2, 1)$  and  $k = \xi n$ , Stirling's formula gives, uniformly on closed subintervals of  $(1/2, 1)$ ,

$$\binom{n}{k}^2 \binom{2k}{n} \binom{n+k}{k} = n^{-2} \kappa(\xi) e^{n\Phi(\xi)} (1 + O(n^{-1})), \quad (42)$$

where, with  $\text{Ent}(u) = -u \log u - (1-u) \log(1-u)$ ,

$$\Phi(\xi) = 2 \text{Ent}(\xi) + 2\xi \text{Ent}\left(\frac{1}{2\xi}\right) + (1+\xi) \text{Ent}\left(\frac{\xi}{1+\xi}\right), \quad (43)$$

$$\kappa(\xi) = \frac{1}{2\pi^2 \xi(1-\xi)} \sqrt{\frac{1+\xi}{2(2\xi-1)}}. \quad (44)$$

A convenient derivative is

$$\Phi'(\xi) = 2 \log(1-\xi) + \log\left(4 + \frac{4}{\xi}\right) - 2 \log(2\xi-1). \quad (45)$$

Furthermore,

$$\Phi''(\xi) = -\frac{2}{1-\xi} - \frac{1}{\xi(1+\xi)} - \frac{4}{2\xi-1} < 0,$$

so the critical point is the unique global maximum. Direct substitution gives

$$\xi_0 = \frac{4}{5}, \quad e^{\Phi(\xi_0)} = 27, \quad \Phi''(\xi_0) = -\frac{625}{36}, \quad \kappa(\xi_0) = \frac{25\sqrt{6}}{16\pi^2}. \quad (46)$$

The endpoint ranges are exponentially smaller than the saddle range. Replacing the sum by a Riemann sum and applying the Gaussian form of Laplace's method therefore yields

$$\begin{aligned} t_n &\sim 27^n n^{-3/2} \kappa(\xi_0) \sqrt{\frac{2\pi}{-\Phi''(\xi_0)}} \\ &= C_t 27^n n^{-3/2}, \quad C_t = \frac{3\sqrt{3}}{4\pi^{3/2}} = \frac{1}{4} \left(\frac{3}{\pi}\right)^{3/2}. \end{aligned} \quad (47)$$

This agrees with Hirschhorn's theorem, whose further terms begin

$$t_n \sim C_t 27^n n^{-3/2} \left(1 - \frac{65}{144n} + \frac{3865}{41472n^2} + \frac{111727}{17915904n^3} + \cdots\right). \quad (48)$$

The first correction  $-65/144$  is also the case  $m = 2$  of manuscript 58's convolution theorem, (92); the two manuscripts agree.

*Remark 6.1* (Why the square alone does not determine the answer). From  $A^2 = \sum t_n z^n$ , the asymptotic constant of  $t_n$  is not simply the square of the constant of  $a_n$ . Both series converge at  $z = \rho$ , and their dominant singularity is additive rather than multiplicative:  $A = A(\rho) + B\sqrt{1-z/\rho} + \cdots$ . Thus the singular coefficient of  $A^2$  is  $2A(\rho)B$ . The missing datum is the exact boundary value  $A(\rho)$ , which is evaluated next.

## 6.4 The Fricke fixed point

Take

$$\tau_* = \frac{i}{\sqrt{7}}. \quad (49)$$

Dedekind's inversion law [11, Ch. 3]

$$\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau) \quad (50)$$

shows that

$$\eta(i\sqrt{7}) = 7^{-1/4} \eta(i/\sqrt{7}). \quad (51)$$

It follows from (39) that  $w_7(\tau_*) = 1/7$ , hence

$$X_7(\tau_*) = \frac{1/7}{1 + 13/7 + 1} = \frac{1}{27} = \rho. \quad (52)$$

All relevant quantities are positive on the imaginary axis. Therefore (41) gives

$$A(\rho) = z_7(\tau_*) = 3\eta(i/\sqrt{7})\eta(i\sqrt{7}). \quad (53)$$

The remaining task is an exact evaluation of this product.

## 6.5 Chowla–Selberg at discriminant $-7$

Put

$$\tau_0 = \frac{1 + i\sqrt{7}}{2}. \quad (54)$$

The class number of discriminant  $-7$  is one [10]. Specializing the Chowla–Selberg formula in the form proved by van der Poorten and Williams [5] gives

$$|\eta(\tau_0)| = (14\pi)^{-1/4} \left( \frac{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)}{\Gamma(3/7)\Gamma(5/7)\Gamma(6/7)} \right)^{1/4}. \quad (55)$$

Reflection [13, Ch. 5] and the classical trigonometric product

$$\sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{8} \quad (56)$$

imply

$$\mathcal{G}_7 \Gamma(3/7)\Gamma(5/7)\Gamma(6/7) = \frac{8\pi^3}{\sqrt{7}}.$$

Consequently (55) simplifies to

$$|\eta(\tau_0)| = \frac{\sqrt{\mathcal{G}_7}}{2 \cdot 7^{1/8} \pi}. \quad (57)$$

We still need to move from  $\tau_0$  to  $i\sqrt{7}$ . Introduce the Weber function

$$\mathfrak{f}_2(\tau) = \sqrt{2} \frac{\eta(2\tau)}{\eta(\tau)}, \quad j(\tau) = \frac{(\mathfrak{f}_2(\tau)^{24} + 16)^3}{\mathfrak{f}_2(\tau)^{24}}. \quad (58)$$

The Hilbert class polynomial of discriminant  $-7$  is  $X + 3375$  [10], so  $j(\tau_0) = -3375$ . Setting  $u = \mathfrak{f}_2(\tau_0)^{24}$  gives

$$(u + 16)^3 + 3375u = (u + 1)(u^2 + 47u + 4096) = 0. \quad (59)$$

Here the nome is  $e^{2\pi i \tau_0} = -e^{-\pi\sqrt{7}}$ , and the product definition of  $\eta$  shows that  $u$  is a negative real number. The two roots of the quadratic factor in (59) are nonreal. Hence  $u = -1$  and  $|\mathfrak{f}_2(\tau_0)| = 1$ . Since  $2\tau_0 = 1 + i\sqrt{7}$  and  $|\eta(\tau + 1)| = |\eta(\tau)|$ , we obtain

$$\eta(i\sqrt{7}) = \frac{|\eta(\tau_0)|}{\sqrt{2}} = \frac{\sqrt{\mathcal{G}_7}}{2\sqrt{2}7^{1/8}\pi}. \quad (60)$$

Combining (51) and (60),

$$\eta(i/\sqrt{7})\eta(i\sqrt{7}) = 7^{1/4}\eta(i\sqrt{7})^2 = \frac{\mathcal{G}_7}{8\pi^2}. \quad (61)$$

Finally, (53) becomes

$$A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}. \quad (62)$$

This is the second proof of  $A(\rho) = A_\star$  in Theorem 2.1.

## 6.6 A second exact series from the functional equation

The square-root generating function satisfies

$$\frac{1}{1+4y}A\left(\frac{y}{(1+4y)^3}\right) = \frac{1}{1+2y}A\left(\frac{y^2}{(1+2y)^3}\right). \quad (63)$$

This follows either from O'Brien's modular equation or by taking the branch equal to 1 at the origin in the corresponding identity for  $A^2$ . At  $y = 1/8$ , the two arguments are  $1/27$  and  $1/125$ , and (63) reads

$$\frac{2}{3}A(1/27) = \frac{4}{5}A(1/125).$$

Together with (62), this proves the second identity of Theorem 2.2.

[Merge note, batch 7302: Provenance of (63). Manuscript 56 derives it from O'Brien's modular equation (or from the identity for  $A^2$ ) and uses it only here. Manuscript 58 mentions, without using it, a functional equation added to the OEIS entry in 2026 (August 2026, in its source audit) as a possible input for the Lucas congruences (Section 13.6); the dossier of the intake identifies it with (63). The intake checked (63) as an identity of power series in  $y$  through  $y^{24}$ , and the value  $A(1/125) = 5\mathcal{G}_7/(16\pi^2)$  by direct summation, to  $3 \times 10^{-61}$ . The second identity of Theorem 2.2 therefore rests on the functional equation, taken as an input from O'Brien's modular equation (the finite check is not a proof), as a power-series identity continued along  $0 \leq y \leq 1/8$ , where both arguments increase within  $[0, \rho]$  and reach  $1/27$  and  $1/125$  at  $y = 1/8$ .]

## 6.7 The singular connection coefficient

Because the exponents at  $s = 0$  are 0 and  $1/2$ , the local form of  $A$  is (19),  $A(z) = U(s) + B\sqrt{s}V(s)$ , with  $U(0) = A(\rho)$  and  $V(0) = 1$ . The coefficient  $B$  is nonzero: otherwise  $A$ , and hence  $A^2$ , would be analytic at  $1/27$ , contradicting (47).

Squaring (19),

$$A(z)^2 = A(\rho)^2 + 2A(\rho)Bs^{1/2} + O(s). \quad (64)$$

The transfer formula (73) at exponent  $1/2$  [12, Ch. VI],

$$[z^n](1 - 27z)^{1/2} \sim -\frac{27^n}{2\sqrt{\pi}n^{3/2}}, \quad (65)$$

then shows that the constant in (47) satisfies

$$C_t = -\frac{2A(\rho)B}{2\sqrt{\pi}} = -\frac{A(\rho)B}{\sqrt{\pi}}. \quad (66)$$

(The transfer from the local expansion of  $A^2$  to  $t_n$  uses the same dented-disk continuation as in Section 3; see the merge note in Section 7.2.) Using (47) and (62),

$$B = -\frac{2\pi\sqrt{3}}{\mathcal{G}_7}. \quad (67)$$

Applying (65) directly to  $A$  gives

$$C = -\frac{B}{2\sqrt{\pi}} = \frac{\sqrt{3\pi}}{\mathcal{G}_7},$$

which is the second proof of the leading assertion of Theorem 2.1. In manuscript 58's terms, (66) is  $C_t = 2CA_\star$ , the case  $m = 2$  of (90); manuscript 56 obtains the product  $CA_\star$  from the known square, manuscript 58 from Ramanujan's series (32).

**Exact local connection formula** (manuscript 56). At the dominant singularity,

$$A(z) = \frac{3\mathcal{G}_7}{8\pi^2} - \frac{2\pi\sqrt{3}}{\mathcal{G}_7}\sqrt{1-27z} + O(1-27z). \quad (68)$$

The first constant is modular/CM; the second is obtained by matching the known square  $A^2$  to the first. Formula (3) is precisely the transfer of the second constant.

Manuscript 58 obtains the same singular coefficient in hypergeometric form, (28), and the same value through  $B = -2\sqrt{\pi}C$ .

## 7 All orders: existence, remainders, and coefficients

### 7.1 The convergent local expansion

Write  $V(s) = \sum_{j \geq 0} v_j s^j$ ,  $v_0 = 1$ ,  $v_{-1} = 0$ . Substitution of  $s^{1/2}V(s)$  into (18) gives

$$\begin{aligned} 28j(j + \tfrac{1}{2})v_j &= \left(29(j - \tfrac{1}{2})^2 + 14(j - \tfrac{1}{2}) + \tfrac{20}{9}\right)v_{j-1} \\ &\quad - \left((j - \tfrac{3}{2})^2 + (j - \tfrac{3}{2}) + \tfrac{2}{9}\right)v_{j-2}, \quad j \geq 1. \end{aligned} \quad (69)$$

Thus all  $v_j$  are rational, with  $v_1 = 593/1512$ . Frobenius theory makes this a *convergent local* power series, not merely a formal solution at infinity.

Manuscript 56 writes the same recurrence with integer coefficients,

$$504(m+1)(2m+3)v_{m+1} - (1044m^2 + 1548m + 593)v_m + (36m^2 - 1)v_{m-1} = 0, \quad (70)$$

and lists

$$\begin{aligned} v_1 &= \frac{593}{1512}, & v_2 &= \frac{7493}{31104}, & v_3 &= \frac{400316059}{2304433152}, \\ v_4 &= \frac{5678514224315}{41811635109888}, & v_5 &= \frac{25841240646919967}{231803705049219072}. \end{aligned} \quad (71)$$

(The two recurrences give the same  $v_j$ ; the intake checked  $j \leq 9$  exactly.) Thus a more detailed version of (68) is

$$A(z) = U(1 - 27z) - \frac{2\pi\sqrt{3}}{\mathcal{G}_7}(1 - 27z)^{1/2} \left( 1 + \frac{593}{1512}(1 - 27z) + \frac{7493}{31104}(1 - 27z)^2 + \dots \right). \quad (72)$$

## 7.2 Transfer to coefficients

For nonintegral  $\alpha$ , the exact coefficient formula is

$$[z^n](1 - 27z)^\alpha = 27^n \frac{\Gamma(n - \alpha)}{\Gamma(-\alpha)\Gamma(n + 1)}. \quad (73)$$

Apply it to  $B \sum_{j=0}^M v_j (1 - 27z)^{j+1/2}$  and expand each gamma ratio [13, §5.11]. Singularity analysis in the dented disk from Section 3 transfers the remainder. The locally analytic part contributes no algebraic terms from this singularity. Equivalently one may subtract arbitrarily many Taylor terms of  $U$  before applying the transfer theorem. The first omitted nonintegral power is  $(1 - 27z)^{M+3/2}$ , and its coefficient is  $O(27^n n^{-M-5/2})$ . This proves (4) for every fixed  $M$ .

[Merge note, batch 7302: Manuscript 56's version of this step reads: "The gamma quotient has a complete expansion in descending powers of  $n$ . Combining the Frobenius series and (73) therefore gives the expansion. The analytic branch contributes only an exponentially smaller term after the local singular part is removed, because the next finite singularity is at  $z = -1$ ." It does not state the continuation domain. The hypothesis singularity analysis needs, continuation to a dented disk of radius greater than  $\rho$  with no other singularity on  $|z| = \rho$ , is manuscript 58's (Section 3); this report relies on it for both proofs.]

The first correction is transparent: the gamma ratio for  $\alpha = 1/2$  contributes  $3/(8n)$ ; the  $s^{3/2}$  term contributes  $-3v_1/(2n)$ , relative to the leading term. Hence

$$c_1 = \frac{3}{8} - \frac{3}{2} \frac{593}{1512} = -\frac{215}{1008}.$$

No numerical fitting of the coefficients is needed.

## 7.3 A triangular recurrence at infinity

For efficient computation insert  $u_n = r^n n^{-3/2} S(1/n)$  in the recurrence, with  $r \in \{27, -1\}$ . The corresponding formal operator is

$$\begin{aligned} \mathcal{E}_r[S](t) &= r(1+t)^{1/2} S\left(\frac{t}{1+t}\right) - (26 + 13t + 2t^2)S(t) \\ &\quad - \frac{27 - 27t + 6t^2}{r} (1-t)^{-3/2} S\left(\frac{t}{1-t}\right). \end{aligned} \quad (74)$$

The characteristic equation is  $r^2 - 26r - 27 = 0$ , with roots  $27, -1$ .

**Proposition 7.1** (Exact coefficient algorithm). *For either root there is a unique formal series  $S_r(t) = 1 + \sum_{j \geq 1} \gamma_j t^j$  with  $\mathcal{E}_r[S_r] = 0$ . Writing  $S_{m-1} = \sum_{j=0}^{m-1} \gamma_j t^j$ , one has*

$$\gamma_m = \frac{[t^{m+1}]\mathcal{E}_r[S_{m-1}](t)}{m(r + 27/r)}, \quad m \geq 1. \quad (75)$$

All coefficients are rational; the denominators in the recursion are  $28m$  and  $-28m$ , respectively.

*Proof.* A newly inserted  $\gamma_m t^m$  contributes zero in order  $m$ , by the characteristic equation. Its coefficient in order  $m + 1$  is

$$r\left(\frac{1}{2} - m\right) - 13 - \frac{27}{r}\left(\frac{1}{2} + m\right) = -m(r + 27/r).$$

This is nonzero, giving the assertion inductively. At the first two orders the normalization is compatible with both characteristic roots.  $\square$

For  $r = 27$ , the analytic existence proof above identifies this formal series with the asymptotic series of  $a_n/C$ . For  $r = -1$ , Section 9 will construct a genuine solution realizing the series. Formal coefficient generation alone is not used to assert existence of an asymptotic expansion.

Manuscript 56 derives the case  $r = 27$  independently: with  $S_{27} = \mathcal{A}$  in its notation,  $\mathcal{E}_{27}[S] = 0$  is its formal identity, since  $(27 - 27t + 6t^2)/27 = 1 - t + \frac{2}{9}t^2$  (Appendix B). It records the next five coefficients beyond (5):

$$\begin{aligned} c_6 &= \frac{5073267927236405}{3082688233381822464}, \\ c_7 &= \frac{141980078251330365605}{21751448174742139305984}, \\ c_8 &= \frac{1180675715260847223544061}{175403678081120611363454976}, \\ c_9 &= -\frac{541114776511779121907204365}{227323166793132312327037648896}, \\ c_{10} &= -\frac{10501970724197421668190580597385}{458283504254954741651307900174336}. \end{aligned} \quad (76)$$

These agree with manuscript 58's recorded coefficients, which continue to  $c_{12}$  in `data/58-gamma-verification.json`. The existence of the local convergent Frobenius series, together with standard singularity analysis, upgrades the formal computation to the error statement in Theorem 2.1 for every fixed order.

## 7.4 Logarithms, ratios, and convexity

Let  $\log S_{27}(t) = \sum_{j \geq 1} \ell_j t^j$ . Then

$$\log a_n = n \log 27 - \frac{3}{2} \log n + \log C + \sum_{j=1}^M \frac{\ell_j}{n^j} + O_M(n^{-M-1}), \quad (77)$$

with

$$\ell_1 = -\frac{215}{1008}, \quad \ell_2 = -\frac{85}{3136}, \quad \ell_3 = \frac{99905}{85349376}, \quad \ell_4 = \frac{1657}{2458624}. \quad (78)$$

In particular,

$$\frac{a_{n+1}}{a_n} = 27 \left( 1 - \frac{3}{2n} + \frac{2105}{1008n^2} + O(n^{-3}) \right). \quad (79)$$

(Corollary 2.3, from manuscript 56, adds the terms in  $n^{-3}$  and  $n^{-4}$ .) The sequence is strictly increasing, since the coefficient of  $a_n$  in (1) already exceeds 1 for  $n \geq 1$ , and  $a_1 > a_0$ . Taking sufficiently many terms before forming second differences gives

$$\log a_{n+1} - 2 \log a_n + \log a_{n-1} = \frac{3}{2n^2} + O(n^{-3}) > 0$$

for sufficiently large  $n$ . Thus it is eventually strictly log-convex. No explicit smallest threshold is asserted.

## 8 A continued fraction and exact rational bracketing

### 8.1 The discrete Wronskian

Let  $b_n$  be as in Theorem 2.6 and set  $W_n = a_n b_{n+1} - a_{n+1} b_n$ . Eliminating the two forward terms gives

$$W_n = -\frac{3(3n-1)(3n-2)}{(n+1)^2} W_{n-1}, \quad W_0 = 1.$$

Grouping factors by residue modulo 3 yields the exact formula

$$W_n = (-1)^n \frac{(3n)!}{n!(n+1)!^2}. \quad (80)$$

Consequently

$$\delta_n := \frac{b_{n+1}}{a_{n+1}} - \frac{b_n}{a_n} = (-1)^n \frac{(3n)!}{n!(n+1)!^2 a_n a_{n+1}}. \quad (81)$$

Its magnitude decreases strictly because

$$\left| \frac{\delta_{n+1}}{\delta_n} \right| = \frac{3(3n+2)(3n+1)a_n}{(n+2)^2 a_{n+2}} < 1.$$

The numerator is one positive summand in the recurrence for the denominator. The ratio tends to  $1/27$  by Theorem 2.1. Thus  $\sum \delta_n$  converges absolutely, and its alternating tails prove the existence of  $L$  and the brackets (13). Even approximants increase and odd approximants decrease; consecutive ones enclose  $L$  with the exact width

$$\left| \frac{b_{n+1}}{a_{n+1}} - \frac{b_n}{a_n} \right| = \frac{(3n)!}{n!(n+1)!^2 a_n a_{n+1}}. \quad (82)$$

No unknown constants enter this stopping criterion.

### 8.2 Polynomial continued-fraction coefficients

The rescaled sequences  $\mathcal{H}_n = (n!)^2 a_n$  and  $\mathcal{K}_n = (n!)^2 b_n$  satisfy

$$u_{n+1} = P_n u_n + Q_n u_{n-1}, \quad P_n = 26n^2 + 13n + 2, \quad Q_n = 3n^2(3n-1)(3n-2), \quad (83)$$

with initial pairs  $(1, 2)$  and  $(0, 1)$ . Their integrality follows directly from this recurrence, independently of integrality of the unscaled sequence. The continuant recurrence gives

$$L = \frac{1}{2 + \frac{6}{41 + \frac{240}{132 + \frac{1512}{275 + \ddots}}}}. \quad (84)$$

The convergent with  $n$  denominator levels is  $\mathcal{K}_n/\mathcal{H}_n = b_n/a_n$ .

*Remark 8.1* (The initial relation matters). The companion satisfies the recurrence only for  $n \geq 1$ ;  $b_1 = 2b_0$  is false. Its generating function  $\mathcal{B}(z) = \sum b_n z^n$  therefore satisfies (17) with right side 1, not 0. The same holds for  $\mathcal{B} - LA$ . These must not be called second analytic solutions of the homogeneous equation at 0. The two-dimensional solution space here consists of sequences satisfying the recurrence for  $n \geq 1$ .

## 9 The recessive scale and its full expansion

Stirling's formula in (80) gives

$$W_n = (-1)^n \frac{\sqrt{3}}{2\pi} \frac{27^n}{n^3} (1 + O(n^{-1})). \quad (85)$$

Together with the dominant asymptotic this yields

$$\delta_n = (-1)^n \frac{\sqrt{3}}{54\pi C^2} 27^{-n} (1 + O(n^{-1})). \quad (86)$$

The exact tail identities are

$$L - \frac{b_n}{a_n} = \sum_{j \geq 0} \delta_{n+j}, \quad e_n = -a_n \sum_{j \geq 0} \delta_{n+j}. \quad (87)$$

Since  $\sum_{j \geq 0} (-1/27)^j = 27/28$ , the leading terms of (16) and (14) follow. The recessive amplitude can also be written

$$D_{\min} = \frac{\sqrt{3}}{56\pi C} = \frac{\mathcal{G}_7}{56\pi^{3/2}}. \quad (88)$$

### 9.1 A remainder proof for the geometric tail

The full Stirling expansion and Theorem 2.1 imply, for each fixed  $M$ ,

$$\delta_n = (-1)^n K_0 27^{-n} \left( \sum_{k=0}^M \eta_k n^{-k} + O_M(n^{-M-1}) \right), \quad K_0 = \frac{\sqrt{3}}{54\pi C^2}, \quad \eta_0 = 1,$$

where all  $\eta_k$  are rational. The error bound is valid for all sufficiently large integer arguments, hence also at  $n + j$  uniformly in  $j \geq 0$ , with a bound by a constant times  $27^{-j} n^{-M-1}$  after factoring out  $27^{-n}$ . Expand  $(n + j)^{-k}$  for  $0 \leq j \leq n/2$  by Taylor's formula, retaining its bounded remainder. The range  $j > n/2$  is exponentially small, even with any fixed polynomial weight. Every coefficient involves moments

$$\sum_{j \geq 0} j^h (-1/27)^j,$$

which are rational: apply  $(q d/dq)^h$  to  $1/(1-q)$  and set  $q = -1/27$ . Thus the tail in (87) has an expansion to every fixed order with a controlled error and rational normalized coefficients. Multiplication by the expansion of  $a_n$  proves existence of (14). Since  $e_n$  satisfies the recurrence, Proposition 7.1 with  $r = -1$  proves that its coefficients are precisely the  $d_j$  already given. This completes the proof of Theorem 2.6.

## 9.2 What the two-scale expansion does and does not say

Because  $W_n \neq 0$ ,  $a_n$  and  $e_n$  form an exact basis of the recurrence solutions for  $n \geq 1$ . Every solution has a unique form  $u_n = \alpha a_n + \beta e_n$ . In this specified basis its two asymptotic scales are

$$u_n \sim \alpha C 27^n n^{-3/2} \sum_{j \geq 0} c_j n^{-j} + \beta (-1)^{n+1} D_{\min} n^{-3/2} \sum_{j \geq 0} d_j n^{-j}. \quad (89)$$

Each exact basis sequence has its own fixed-order expansion. A finite truncation of the dominant expansion generally has an error far larger than the entire recessive term. Equation (89) is therefore not a claim that two finite sums resolve exponentially small effects in a generic solution. A canonical exponentially improved expansion of  $a_n$  alone would require additional summation or Stokes data; that question is left open.

## 10 Convolution powers and A183204

For a fixed positive integer  $m$ , let

$$a_n^{(m)} = [z^n] A(z)^m = \sum_{n_1 + \dots + n_m = n} a_{n_1} \cdots a_{n_m}.$$

The exact boundary value controls this family as well as the original sequence.

**Theorem 10.1** (Convolution asymptotics; manuscript 58). *For every fixed  $m \geq 1$ ,*

$$a_n^{(m)} = m A_\star^{m-1} C \frac{27^n}{n^{3/2}} \left( 1 + \frac{c_1^{(m)}}{n} + O_m(n^{-2}) \right), \quad (90)$$

where

$$c_1^{(m)} = -\frac{215}{1008} - \frac{5}{21}(m-1) - \pi \frac{C^2}{A_\star^2} (m-1)(m-2). \quad (91)$$

*There is a computable expansion to every fixed order, and all its normalized coefficients lie in  $\mathbb{Q}(\pi C^2/A_\star^2)$ .*

*Proof.* The analytic part of (19) satisfies

$$U(s) = A_\star \left( 1 + \frac{10}{63}s + O(s^2) \right),$$

as the constant coefficient of (18) shows. Its subsequent normalized coefficients are rational. The singular part is  $B\sqrt{s}(1 + v_1 s + \dots)$ , with  $B = -2\sqrt{\pi}C$  and  $v_1 = 593/1512$ . In the  $m$ th power the coefficient of  $\sqrt{s}$  is  $m A_\star^{m-1} B$ . Relative to that coefficient, the coefficient of  $s^{3/2}$  is

$$v_1 + (m-1) \frac{10}{63} + \frac{(m-1)(m-2)}{6} \frac{B^2}{A_\star^2}.$$

Transfer as in Section 7 gives (91). At higher orders only odd powers of  $B$  multiply half-integral powers of  $s$ . After division by the leading singular coefficient, the remaining expressions are polynomials in  $B^2/A_\star^2 = 4\pi C^2/A_\star^2$  with rational coefficients. The convergent Puiseux expansion and the same dented domain provide the fixed-order remainder estimates.  $\square$

For  $m = 2$  this is A183204, and the first two terms simplify to

$$[z^n]A(z)^2 = \frac{3\sqrt{3}}{4\pi^{3/2}} \frac{27^n}{n^{3/2}} \left(1 - \frac{65}{144n} + O(n^{-2})\right). \quad (92)$$

The leading formula is already recorded in its OEIS entry [2]; it is recovered here, not claimed as new. The cancellation of gamma factors at this convolution power is a useful check on the evaluation of  $C$ . (Both terms are Hirschhorn's [4], as quoted by manuscript 56 in (48), which runs on to the  $n^{-3}$  term. In the second proof of Section 6, the logic is reversed: the leading constant of (92) is the input from which  $C$  is extracted.)

For an algebraic check of the generating-function identification, the symmetric-square equation of (17) is

$$\begin{aligned} &(-z^2 + 26z^3 + 27z^4)Y''' + (-3z + 117z^2 + 162z^3)Y'' \\ &+ (-1 + 86z + 186z^2)Y' + (4 + 24z)Y = 0. \end{aligned} \quad (93)$$

This is the equation recorded for A183204, and its normalized analytic solution is unique. The program verifies this symmetric-square identity and checks the convolution against

$$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{n} \binom{2n-k}{n}$$

for  $0 \leq n \leq 40$ . The finite check supplements the differential-equation argument; it does not replace it.

## 11 Inversion and the integer-rounding boundary

[Merge note, batch 7302: Both manuscripts invert the asymptotic by Lambert's  $W_{-1}$ . The core and its triangular corrections are the instance  $a = \lambda = \log 27$ ,  $\beta_{\text{CTI}} = -3/2$ ,  $q_j^{\text{CTI}} = \ell_j$  of Theorem `t2:thm:balanced-inverse` of the repository's volume *Combinatorial Transseries and Their Inverses* [16]: the coefficients  $q_1, q_2$  of (97) below are that theorem's  $d_1, d_2$  (checked by hand at intake). Neither manuscript cites it, and this report claims no novelty for the inverse expansion. The rounding proposition, Proposition 11.2, is the sequence-specific part; its conclusion has the form of the separation condition (part 2) of Theorem `p0:thm:staircase` in the repository's canonical volume *Transseries and Inversion* [17].]

Put  $\lambda = \log 27$  and  $\beta = 3/2$ . The leading smooth model is  $Y = Ce^{\lambda x}x^{-\beta}$ . Its large positive solution is

$$x_0(Y) = -\frac{\beta}{\lambda} W_{-1} \left( -\frac{\lambda}{\beta} (C/Y)^{1/\beta} \right), \quad (94)$$

where  $W_{-1}$  is the real negative branch of Lambert's function. The principal branch selects the small solution and is inappropriate for inversion of the large- $n$  sequence [13, §4.13]. (Manuscript 56 derives the same formula, raising the envelope to the power  $-2/3$ .)

### 11.1 A triangular inverse expansion

Seek

$$x_M(Y) = x_0 + \sum_{j=1}^M \frac{q_j}{x_0^j}. \quad (95)$$

Writing  $\delta = x - x_0$ , the formal equation determining the coefficients is

$$\lambda\delta - \beta \log(1 + \delta/x_0) + \sum_{k \geq 1} \ell_k (x_0 + \delta)^{-k} = 0. \quad (96)$$

The newly introduced coefficient  $q_j$  has multiplier  $\lambda$ , so this is triangular at every order. The first coefficients are

$$\begin{aligned} q_1 &= -\frac{\ell_1}{\lambda} = \frac{215}{1008\lambda}, \\ q_2 &= \frac{\beta q_1 - \ell_2}{\lambda} = \frac{215}{672\lambda^2} + \frac{85}{3136\lambda}, \\ q_3 &= \frac{\beta q_2 + \ell_1 q_1 - \ell_3}{\lambda}. \end{aligned} \quad (97)$$

All  $q_j$  belong to  $\mathbb{Q}(\lambda)$ .

An expansion involving only logarithms follows by putting  $X = \log(Y/C)$ :

$$\begin{aligned} x(Y) &= \frac{1}{\lambda} \left\{ X + \beta \log \frac{X}{\lambda} + \frac{\beta^2 \log(X/\lambda) - \lambda \ell_1}{X} \right\} \\ &\quad + O\left(\frac{(\log X)^2}{X^2}\right). \end{aligned} \quad (98)$$

Here  $x(Y)$  denotes the asymptotic inverse scale, not a previously specified analytic interpolation of the integer sequence. Formula (94) is often more accurate numerically because it retains the entire leading model instead of truncating its logarithmic inversion.

*Remark 11.1* (Manuscript 56's form of the inverse). Expanding  $W_{-1}$  at  $0^-$ , manuscript 56 records

$$x_0(Y) = \frac{1}{\lambda} \left( \log Y + \frac{3}{2} \log \log Y - \frac{3}{2} \log \lambda - \log C \right) + O\left(\frac{\log \log Y}{\log Y}\right), \quad (99)$$

consistent with the first two terms of (98). It notes that the rational correction series can be inserted and reverted recursively to produce an arbitrarily long expansion in  $1/\log Y$  with nested powers of  $\log \log Y$ , and that for the discrete inverse  $N(Y)$  below, formula (94), “rounded and corrected by one or two exact recurrence evaluations, gives an efficient certified locator”. [*Merge note, batch 7302*: The last statement is a practical recipe, not a theorem: a *certified* locator needs explicit constants  $K_M, Y_M$  in Proposition 11.2, which neither manuscript supplies (manuscript 58 lists them as a research direction, Section 13.1). Checking a candidate  $n$  by exact evaluation of  $a_{n-1} < Y \leq a_n$  is of course rigorous.]

## 11.2 A rigorous discrete statement

Define

$$N(Y) = \min\{n \geq 0 : a_n \geq Y\}.$$

The sequence is strictly increasing and unbounded, so this is well-defined for  $Y > 1$ . An inverse approximation cannot uniformly remove the ambiguity when its value is exceptionally close to an integer.

**Proposition 11.2** (Rounding brackets). *For every fixed  $M \geq 0$  there exist  $K_M > 0$  and  $Y_M$  such that, for  $Y \geq Y_M$ ,*

$$\lceil x_M - K_M x_M^{-M-1} \rceil \leq N(Y) \leq \lceil x_M + K_M x_M^{-M-1} \rceil. \quad (100)$$

*When the two ceilings coincide, they determine the discrete inverse exactly. For  $M = 0$ ,  $x_M$  means  $x_0$ .*

*Proof.* Use the smooth finite model

$$g_M(x) = \log C + \lambda x - \beta \log x + \sum_{k=1}^M \ell_k x^{-k}.$$

The all-orders theorem gives  $\log a_n = g_M(n) + O_M(n^{-M-1})$ . Also  $g'_M(x) = \lambda + O(x^{-1})$ , so it is bounded away from zero for large  $x$ . Let  $t_M$  be the large solution of  $g_M(t_M) = \log Y$ . Formal substitution of (95) into  $g_M$ , with ordinary finite Taylor remainders, gives  $g_M(x_M) - \log Y = O_M(x_M^{-M-1})$ . The mean value theorem therefore implies  $t_M - x_M = O_M(x_M^{-M-1})$ .

For integers near  $t_M$ , comparison of  $g_M(n)$  with  $g_M(t_M)$ , using its derivative lower bound and the error in  $\log a_n$ , shows that integers sufficiently below  $t_M$  cannot reach  $Y$ , whereas integers sufficiently above it do reach  $Y$ . Enlarge the constant to absorb  $t_M - x_M$ , and use monotonicity of  $a_n$  for all remaining integers. This gives the two ceilings in (100).  $\square$

The proposition asserts the existence of fixed-order constants, not explicit numerical values of  $K_M$  or  $Y_M$ . Computing certified, practical values is one of the research directions below. No claim that  $N(Y) = \lceil x_M(Y) \rceil$  holds uniformly is made.

## 12 Reproducible computations and certified constants

[Merge note, batch 7302: File names in this section are the shipped ones. Manuscript 58 named its README, SOURCE\_AUDIT.md and data/verification.json; they are this report's README.md, 58-gamma-SOURCE\_AUDIT.md and data/58-gamma-verification.json.]

Build and run instructions are in the accompanying README.md; manuscript 58's source audit is in 58-gamma-SOURCE\_AUDIT.md. The PDF can be built without running Python, since its numerical tables are part of the TeX source. Manuscript 58's program (code/58-gamma-verify.py) uses exact integers and rational functions for its algebraic checks. Arbitrary-precision floating-point arithmetic is used separately for numerical diagnostics. It requires only Python, SymPy, and mpmath, and makes no network requests.

### 12.1 Certificates and reproducible output

The following assertions were executed successfully: eight rational identities for the two pullbacks and two symmetric squares; coefficient recurrences through order 12 for both characteristic roots; exact recurrence evaluation through  $n = 1000$ ; 100 discrete Wronskian identities; and 41 comparisons with the binomial sum for A183204. The division in the  $a_n$  recurrence is exact at every tested index. This last computation is a finite integrality check, not an independent proof of integrality for all indices.

The constants are

$$\begin{aligned} C &= 0.09552230526812671465130791078702962567279466665100\dots, \\ A_\star &= 1.22113109746756666928422458568599346237582705637935\dots, \\ L &= 0.46721176888437312394211005141309126172410854995971\dots, \\ D_{\min} &= 0.10306659600794607491874204424062776206902297866964\dots \end{aligned}$$

The first displayed value is not inferred by integer-relation searching: its gamma formula is proved in Section 5 (and again in Section 6), and the following interval method supplies independent certified decimal evaluation.

### 12.2 A rational enclosure for the asymptotic constant

Write  $F(q_0) = \sum_{j \geq 0} h_j$ . Its positive terms satisfy

$$\frac{h_{j+1}}{h_j} = q_0 \frac{(12j+1)(12j+5)}{144(j+1)^2} < q_0 < 1.$$

Hence, for  $S_N = \sum_{j=0}^{N-1} h_j$ ,

$$S_N \leq F(q_0) \leq S_N + \frac{h_N}{1 - q_0}. \quad (101)$$

Every quantity in this enclosure is rational. The implementation bounds  $\pi$  using Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and exact alternating-series bounds. Square and fourth roots are bracketed by rational numbers using integer root extraction on a fixed decimal grid. Combining positive upper and lower bounds in (32) gives a rational enclosure for  $C$  of width less than  $10^{-120}$ . Outward rounding produces the following interval of width  $10^{-100}$ ; each row continues on its second line.

Lower: 0.09552230526812671465130791078702962567279466665100  
 71798669948234917659264524531457675812093972181446  
 Upper: 0.09552230526812671465130791078702962567279466665100  
 71798669948234917659264524531457675812093972181447

For  $L$  the enclosure uses no transcendental functions at all:  $b_{90}/a_{90} < L < b_{91}/a_{91}$ , with exact rational endpoints whose difference is less than  $10^{-125}$ . The full endpoints and their 100-place decimal enclosures are stored in `data/58-gamma-verification.json`. The JSON also records every coefficient through order 12 and the zero values of all eight rational certificates.

### 12.3 Numerical diagnostics

For Table 2, the relative error is

$$\frac{C 27^n n^{-3/2} \sum_{j=0}^M c_j n^{-j}}{a_n} - 1.$$

The values are rounded floating-point diagnostics, not new rigorous upper bounds on the asymptotic remainder. All were computed with 190 decimal digits of working precision.

Table 2: Dominant expansion: signed relative errors (manuscript 58).

| $n$  | $M = 0$                | $M = 3$                  | $M = 6$                  | $M = 12$                |
|------|------------------------|--------------------------|--------------------------|-------------------------|
| 20   | $1.079 \times 10^{-2}$ | $-1.143 \times 10^{-9}$  | $-5.413 \times 10^{-12}$ | $2.818 \times 10^{-19}$ |
| 100  | $2.138 \times 10^{-3}$ | $-2.449 \times 10^{-12}$ | $-6.609 \times 10^{-17}$ | $6.083 \times 10^{-28}$ |
| 1000 | $2.133 \times 10^{-4}$ | $-2.597 \times 10^{-16}$ | $-6.536 \times 10^{-24}$ | $6.727 \times 10^{-41}$ |

The normalized recessive error in Table 3 is  $\sum_{j=0}^M d_j n^{-j} - e_n / [(-1)^{n+1} D_{\min} n^{-3/2}]$ . The value of  $L$  used in this diagnostic is the midpoint of the exact  $r_{150}, r_{151}$  enclosure, where  $r_n = b_n/a_n$ . Its error remains negligible after multiplication by  $a_n$  at the displayed indices.

Table 3: Recessive expansion: signed normalized errors (manuscript 58).

| $n$ | $M = 0$                | $M = 3$                 | $M = 6$                  |
|-----|------------------------|-------------------------|--------------------------|
| 10  | $4.910 \times 10^{-2}$ | $1.721 \times 10^{-6}$  | $-1.532 \times 10^{-9}$  |
| 30  | $1.676 \times 10^{-2}$ | $2.361 \times 10^{-8}$  | $-6.532 \times 10^{-13}$ |
| 75  | $6.753 \times 10^{-3}$ | $6.240 \times 10^{-10}$ | $-1.036 \times 10^{-15}$ |

Table 4 tests inversion at  $Y = a_n$ . It reports the signed continuous error  $x_M(a_n) - n$ , not an integer rounding error. The uncorrected Lambert- $W$  scale is already accurate; the algebraic corrections produce the predicted further improvement.

Table 4: Inverse expansion at exact sequence values (manuscript 58).

| $n$  | $x_0(a_n) - n$          | $x_1(a_n) - n$          | $x_3(a_n) - n$           |
|------|-------------------------|-------------------------|--------------------------|
| 20   | $-3.332 \times 10^{-3}$ | $-9.581 \times 10^{-5}$ | $-6.315 \times 10^{-9}$  |
| 100  | $-6.509 \times 10^{-4}$ | $-3.780 \times 10^{-6}$ | $-1.064 \times 10^{-11}$ |
| 1000 | $-6.475 \times 10^{-5}$ | $-3.769 \times 10^{-8}$ | $-1.075 \times 10^{-15}$ |

## 12.4 Manuscript 56's tables and checks

All numerical checks of manuscript 56 were generated from exact integers using (1); no OEIS term table was used after the initial comparison. Define the scaled quantity

$$R_n = \frac{a_n n^{3/2}}{27^n}. \quad (102)$$

Table 5 compares  $R_n$  with the exact constant. The last column subtracts the approximation through  $c_3/n^3$  and divides the residual by  $C$ .

Table 5: Convergence to the exact asymptotic constant (manuscript 56).

| $n$  | $R_n$                 | $R_n/C - 1$                 | residual after $c_3$     |
|------|-----------------------|-----------------------------|--------------------------|
| 10   | 0.0934812234164389792 | $-2.1367594 \times 10^{-2}$ | $1.1427 \times 10^{-8}$  |
| 50   | 0.0951146568282489133 | $-4.2675733 \times 10^{-3}$ | $3.6462 \times 10^{-11}$ |
| 100  | 0.0953185211418410836 | $-2.1333669 \times 10^{-3}$ | $2.4441 \times 10^{-12}$ |
| 500  | 0.0954815550048036949 | $-4.2660469 \times 10^{-4}$ | $4.1263 \times 10^{-15}$ |
| 1000 | 0.0955019305511748892 | $-2.1329800 \times 10^{-4}$ | $2.5960 \times 10^{-16}$ |
| 5000 | 0.0955182303912366763 | $-4.2658904 \times 10^{-5}$ | $4.1754 \times 10^{-19}$ |

The residual column is the  $M = 3$  column of Table 2 with the opposite sign convention: manuscript 56 tabulates  $R_n/C - \sum_{j \leq 3} c_j n^{-j}$ , manuscript 58 tabulates  $C \sum_{j \leq 3} c_j n^{-j}/R_n - 1$ , which is minus the former times  $C/R_n = 1 + O(n^{-1})$ . At  $n = 100$  and  $1000$  the two agree up to sign to the printed digits ( $2.4441 \times 10^{-12}$  against  $-2.449 \times 10^{-12}$ , and  $2.5960 \times 10^{-16}$  against  $-2.597 \times 10^{-16}$ ).

The critical series converges slowly, exactly as Corollary 2.5 predicts. Table 6 gives the exact remainder computed from (62) and its ratio to the leading estimate  $2C/\sqrt{N} + 1$ .

Table 6: Critical-series tail at  $z = 1/27$  (manuscript 56).

| $N$  | $A(1/27) - \sum_{n=0}^N a_n/27^n$ | ratio to $2C/\sqrt{N} + 1$ |
|------|-----------------------------------|----------------------------|
| 10   | 0.058541895880356                 | 1.016314999                |
| 50   | 0.026845518682688                 | 1.003510906                |
| 100  | 0.019043336251750                 | 1.001772100                |
| 500  | 0.008538291869020                 | 1.000357123                |
| 1000 | 0.006039421866043                 | 1.000178732                |
| 5000 | 0.002701605297719                 | 1.000035774                |

Manuscript 56's script (`code/56-exact-verify_a279619.py`; its recorded output is `data/56-exact-verification.txt`) performs the following independent checks:

1. generates  $a_0, \dots, a_N$  as exact Python integers and verifies divisibility by  $(n+1)^2$  at every recurrence step;
2. derives the  $c_j$  by exact symbolic expansion of the formal identity  $\mathcal{E}_{27}[S] = 0$  of (74);
3. derives the  $v_j$  by the exact recurrence (70);
4. evaluates the gamma products and both special sums at high precision;
5. prints Tables 5 and 6;
6. checks the algebraic identities for  $A(1/27)$ ,  $B$ ,  $C$ , and the square-sequence constant  $C_t$ .

Items 4 and 6 are high-precision numerical consistency checks of identities proved in Section 6; they are not proofs.

## 13 Further research questions and topics

The questions below are concrete extensions of the proved results. They are not assertions that every item is globally open: before a priority claim, the relevant modular-form and continued-fraction literature should be checked in greater depth. Sections 13.1–13.12 are manuscript 58's twelve; Section 13.13 lists manuscript 56's ten and records where they meet manuscript 58's.

### 13.1 Explicit remainder constants

Theorems 2.1 and 2.6 give remainders at every fixed order but not practical values of the constants. Construct computable  $K_M, N_M$  such that

$$\left| \frac{a_n n^{3/2}}{C 27^n} - \sum_{j=0}^M c_j n^{-j} \right| \leq K_M n^{-M-1} \quad (n \geq N_M).$$

One route is to bound the analytic Frobenius functions on a specified complex neighborhood and make the transfer contour explicit. A second is to construct upper and lower recurrence solutions from truncated series plus a controlled correction. The latter may give smaller thresholds and could turn the inverse brackets into a practical certified algorithm.

### 13.2 Late-order growth and optimal truncation

Determine the large- $j$  behavior of  $c_j$  and  $d_j$ . The proof of fixed-order asymptotics does not imply convergence of the series in  $1/n$ . A useful target is a sharp factorial-over-power law, with its constant and phase, or a proof of a specified Gevrey bound. This would identify an optimal truncation index and quantify what accuracy is available before the recessive scale becomes relevant.

### 13.3 Canonical exponentially improved asymptotics

The exact basis  $a_n, e_n$  removes ambiguity from (89), but it does not specify a canonical summation of the dominant formal series. Develop a Borel or contour normalization and compute the associated connection data between the singularities  $\rho$  and  $-1$ . The desired result is an error theorem resolving the alternating contribution after optimal truncation, not merely a formal addition of a second exponential scale.

### 13.4 Recognizing the continued-fraction limit

Find an exact special-value expression for  $L$ . The inhomogeneous operator in the remark following (84) provides a starting point: variation of parameters expresses its solutions through integrals involving  $A$  and algebraic functions. The condition that the  $\sqrt{1-27z}$  coefficient vanish selects  $\mathcal{B} - LA$  and hence  $L$ . A successful reduction might involve an elliptic integral, an iterated modular integral, or a special  $L$ -value. No such identification is proved in this report.

### 13.5 Arithmetic strength of the approximants

Does the continued fraction prove irrationality of  $L$ , or a useful irrationality measure? The exponential error  $27^{-n}$  alone is not enough: the natural continuant denominators contain factorial growth. The first step is to analyze the reduced denominators and common factors of  $\mathcal{H}_n$  and  $\mathcal{K}_n$ . Any arithmetic conclusion must compare the error with the actual denominator after cancellation. The general results in [14] are relevant context, but do not automatically apply to the specific limit constructed here.

### 13.6 The arithmetic conjecture in the entry

The OEIS entry also displays Lucas congruences for primes in the specified residue classes modulo 7 [1]. The present asymptotic argument neither proves nor disproves them. A separate project should first audit subsequent work on the Chan–Cooper–Sica conjecture and then exploit the modular parametrization or the functional equation added to OEIS in 2026 (printed as (63)). This is a different target from the asymptotic conjecture resolved here; analytic evidence should not be substituted for an arithmetic proof.

### 13.7 Coefficient fields for convolution powers

Theorem 10.1 bounds the coefficient field by  $\mathbb{Q}(\pi C^2/A_\star^2)$ . Determine exactly which convolution powers and which correction orders eliminate the period ratio. The square is special at its first correction, and its third-order differential equation may expose cancellations more systematically. A symbolic classification would clarify when a convolution inherits purely rational normalized corrections and when new period combinations necessarily appear.

### 13.8 Noninteger powers and growing convolution order

Extend the convolution theorem to a specified branch of  $A(z)^u$  for noninteger real or complex  $u$ . Zero avoidance and possible additional singularities must be addressed rather than assumed. A separate uniform problem lets the integer  $m = m(n)$  grow. The fixed- $m$  proof is then insufficient because powers of the analytic boundary part and the singular part can compete. Determine the applicable uniform ranges and the transition scales from the actual local expansion.

### 13.9 Positive integral representations

Find a direct integral or moment representation for  $a_n$  that makes positivity and possible stronger inequalities transparent. A compact real support reflecting both characteristic roots would be especially informative, but such a representation and positivity of its density are not established here. Potential consequences include explicit log-convexity thresholds, Hankel determinant signs, and alternative error bounds for coefficient asymptotics.

### 13.10 Automated connection-constant recognition

For families of quadratic-coefficient recurrences with two distinct characteristic roots, combine three exact procedures: derive the Fuchsian equation, certify a rational hypergeometric pullback, and test whether a singular endpoint is a known complex-multiplication value. The intended output should include a symbolic certificate and a branch check, not just a high-precision guess. A279619 is a compact test case because the rational parameter turns the dominant endpoint into  $w = 0$ .

### 13.11 A direct modular proof of the product identity

Can the product  $CA_\star = 3\sqrt{3}/(8\pi^{3/2})$  be obtained directly from a single level-seven modular relation? The proof here combines a Ramanujan identity with a period evaluation. A direct connection or Wronskian argument on the modular side may unify the two normalizations and generalize to other Apéry-like equations. Such a proof could also explain which sequences exhibit reciprocal boundary and singular periods. [*Merge note, batch 7302*: Manuscript 56's proof (Section 6) obtains  $CA_\star$  as half of Hirschhorn's constant  $C_t$  of the square, and  $A_\star$  from a single CM eta product. It is modular on the boundary value but still takes the singular coefficient from the separately proved asymptotic of A183204, so the question as posed, a single modular relation giving the product, remains open.]

### 13.12 Machine-checked certificates and analytic infrastructure

A staged Lean development can begin with the recurrence, positivity, integer continuants, and the exact Wronskian. These statements need only algebra and induction. The rational identities in (26) and the triangular coefficient algorithm form a second self-contained layer. A later analytic layer must connect convergent local solutions to coefficient asymptotics, including branch choices and remainder estimates. The complex-multiplication identities should be stated as explicit dependencies until they too are formalized. This separation prevents an exact symbolic certificate from being mistaken for a proof of analytic continuation or of a special value.

### 13.13 Manuscript 56's questions

Manuscript 56 wrote: the exact constant closes one visible gap, but it exposes several deeper ones. Its list is ordered roughly by mathematical importance and by the likelihood that the existing level-seven structure can be used. Each item is followed, where it applies, by its relation to the items above.

**R1. Prove the Lucas-congruence conjecture for A279619.** The OEIS entry conjectures Lucas congruences for primes  $p \equiv 0, 1, 2, 4 \pmod{7}$ . A promising route is to combine a diagonal or constant-term representation with the Cartier operator and Dwork-style unit-root arguments. The exceptional residue

classes should be explained geometrically rather than checked case by case. (Same target as Section 13.6; different suggested route.)

- R2. Find a direct binomial or multisum formula for  $a_n$ .** The square coefficients  $t_n$  have the compact sum (37), while the square-root coefficients are presently most natural through their recurrence or modular expansion. A positive multisum would likely clarify integrality, congruences, and combinatorial meaning simultaneously.
- R3. Determine supercongruences and the underlying motive.** A183204 belongs to the sporadic Apéry-like family [9]. The square-root sequence may encode a lower-weight or half-integral structure. Identify the relevant crystalline Frobenius polynomial and test whether  $a_{mp^r}$  satisfies congruences stronger than Lucas congruence.
- R4. Classify all CM evaluations of  $A(X_7(\tau))$ .** For imaginary quadratic  $\tau$ , Chowla–Selberg expresses eta values in gamma products. Determine which CM points give algebraic  $X_7(\tau)$ , then tabulate exact series evaluations  $\sum a_n X_7(\tau)^n$ . The values at  $1/27$  and  $1/125$  should be the first entries of a larger level-seven atlas.
- R5. Compute exact connection matrices.** Equation (17) has singular points  $0, 1/27, -1$ , and  $\infty$ . The present article computes one connection coefficient between  $0$  and  $1/27$ . Determine the full  $2 \times 2$  connection matrices between Frobenius bases at all singular points, preferably in closed gamma form. (Manuscript 58’s companion adds the exact recessive amplitude  $D_{\min}$  of (88), a connection datum of the recurrence at the root  $-1$ ; the full matrices remain open.)
- R6. Establish global log-convexity or stronger total positivity.** We proved eventual log-convexity. Exact computation suggests stronger positivity phenomena. Seek a Stieltjes-moment representation, a continued fraction, or a production matrix proving log-convexity for all  $n$  and perhaps total positivity of the Hankel matrix  $(a_{i+j})$ . (Overlaps “Positive integral representations” above.)
- R7. Develop rigorous optimal truncation and exponentially small terms.** The algebraic expansion is controlled by the singularity at  $1/27$ . After optimal truncation, the next singularity at  $-1$  should generate an exponentially small, alternating contribution. Determine the associated transseries and Stokes constants. (Same target as manuscript 58’s late-order and exponentially improved items above. Manuscript 58’s Theorem 2.6 constructs the alternating recessive solution of the recurrence exactly; what remains open is the canonical summation and the Stokes data of  $a_n$  itself.)
- R8. Invert the full expansion.** Extend (94) to a systematic inverse transseries for the integer threshold  $N(Y)$ , including rational corrections and a proof of a uniform rounding rule. (Re-scoped: the systematic inverse with rational corrections is (95)–(97), an instance of `t2:thm:balanced-inverse`, and the rounding brackets are Proposition 11.2. A *uniform* rounding rule is not available: Proposition 11.2 decides  $N(Y)$  only when the two ceilings coincide, and manuscript 58 makes no claim that  $N(Y) = \lceil x_M(Y) \rceil$  uniformly. What remains open is explicit  $K_M, Y_M$ , Section 13.1.)
- R9. Formalize the algebraic core.** A realistic Lean project would formalize the recurrence-to-ODE step, the Frobenius coefficient recurrence, the formal identity  $\mathcal{E}_{27}[S] = 0$ , and finite-order transfer algebra. The Dedekind eta and Chowla–Selberg inputs would initially be isolated as analytic theorems, making the boundary between formalized algebra and classical complex analysis explicit. (Same programme as Section 13.12.)
- R10. Audit neighboring level-seven OEIS entries.** The same thesis introduced the quartic-recurrence sequence A279613 and related modular functions. Its oscillatory asymptotic is still listed conjecturally.

The exact-CM/connection strategy used here suggests a systematic attack on its amplitude and phase constants, although the multiple dominant singularities make the problem substantially harder.

## 14 Scope, audits, and conclusion

### 14.1 Manuscript 56's scope and novelty audit

The mathematical proof of Section 6 uses published structural identities, but the closed form (3) is obtained there by combining them in a way not stated in the inspected sources. The audit performed for manuscript 56 covered:

- the current OEIS pages for A279619 and A183204;
- O'Brien's 2016 thesis, including its recurrence, modular parametrization, square relation, and numerical asymptotic work;
- Hirschhorn's proof of the A183204 asymptotic;
- van der Poorten and Williams' explicit eta-value formula;
- targeted web and repository searches for A279619 together with the gamma product or the displayed decimal constant;
- the existing OEIS-asymptotics reports in the ProveIt repository.

No prior exact identification of the A279619 constant was found in that search: the constant appears not to have been recorded in the OEIS entry, O'Brien's thesis, Hirschhorn's paper, or the targeted searches. This is evidence of novelty, not a substitute for a full bibliographic review or peer review, and not a claim that no equivalent formula can exist elsewhere. All theorem statements are supported by the derivations given and by the reproducible checks shipped with the source. Manuscript 58's audit (Section 1 and 58-gamma-SOURCE\_AUDIT.md) reached the same conclusion independently.

### 14.2 Integration roadmap and conclusion

A natural repository layout separates four components: sequence recurrences and exact certificates; local analytic representation; asymptotic coefficient calculus; and inverse expansions. The first and third components can be tested independently of modular-form machinery. The second supplies the analytic existence and connection constants that formal manipulations alone cannot establish. The fourth should retain an explicit distinction between a continuous inverse scale and an integer threshold function.

No existing Lean declaration is assumed to certify the special-function steps of this report. A useful first integration milestone is a machine-checked discrete Wronskian with rational bracketing of  $L$ . The next is a generic formal-series lemma giving (75); both are substantially smaller than a full formalization of the gamma constant. The supplied Python verification programs are a reproducible algebraic audit and a source of test data, not a replacement for that formalization.

The principal outcome is a precise resolution of the asymptotic statement displayed for A279619:

$$a_n \sim \frac{\sqrt{3}\pi}{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)} \frac{27^n}{n^{3/2}}.$$

The same argument supplies the boundary sum, every algebraic correction with a fixed-order error bound, and a useful convolution calculus. The independent companion construction reveals an alternating recessive solution with an exact gamma-product amplitude. Inversion then yields a Lambert- $W$  scale and rigorous rounding brackets. The remaining questions concern sharper effective bounds, canonical exponentially improved expansions, arithmetic properties, and broader families, rather than the existence or value of the leading asymptotic constant proved here.

## 15 Provenance of this report

- **Two manuscripts**, both of batch 73O2 of ProveIt’s incoming-reports intake, both dated October 1, 2026:
  - manuscript 58, *An exact gamma constant for an Apéry-like sequence* (author line “Research report prepared for Vladimir Reshetnikov”; 22-page PDF), archive A279619\_Gamma\_Asymptotics.zip, arrived in commit aa43cc555; pin a5ad4ac2e2d4e642a6c015836a70492d40fcb71 (the repository snapshot it inspected). It is the base text: everything not marked otherwise.
  - manuscript 56, *Exact asymptotics of OEIS A279619: a level-7 CM evaluation, the missing connection constant, and a full coefficient expansion* (author line “A mathematical analysis prepared for Vladimir Reshetnikov”; 15-page PDF), archive A279619\_exact\_asymptotics.zip, arrived in commit 58dbe2cd3; no pin (it cites no repository revision). It contributes the material listed in Section 1.3.
- **Placement:** commit 6e193dd4f staged manuscript 58’s article, README and source audit and both manuscripts’ code and data (prefixes 58-gamma- and 56-exact-); manuscript 56’s article and README and both PDFs were not staged and survive in their arrival commits.
- **Independence.** Manuscript 58’s pin (12:35 PDT) is earlier than manuscript 56’s arrival (12:56 PDT); manuscript 56 cites no revision. Neither saw the other. Their texts share formulas, the opening terms and the O’Brien bibliography entry, but no prose.
- **Where the merge chose.** (i) Manuscript 58’s analytic justification (the dented disk) is used for both proofs; manuscript 56’s transfer step is kept verbatim in a merge note (Section 7.2). (ii) Results proved in both are printed once in manuscript 58’s wording (the main theorem,  $A(\rho)$ , the differential equation, the Frobenius recurrence, the formal identity at infinity, the Lambert- $W$  core, the A183204 constant), with manuscript 56’s distinct proofs and forms kept. (iii) Manuscript 56’s notation was rewritten in manuscript 58’s (Section 1.4). (iv) Both abstracts were replaced by one; both title pages by one. (v) Manuscript 56’s reproducibility appendix, which lists its delivery file names, is replaced by the report README. (vi) The two OEIS proposals are both printed (Appendix C); manuscript 56’s gains the credit to O’Brien that manuscript 58’s already had. (vii) The two bibliographies are merged; five entries that manuscript 56 listed but never cited (Cooper, Cox, Apostol, DLMF, Flajolet–Sedgewick) are now cited where they are used.

## A Checking the classical period normalization

The table in Zucker’s singular-value notes [8, Table 1] uses  $K[N] = K(k_N)$  with  $K(k'_N)/K(k_N) = \sqrt{N}$ . At  $N = 7$  it gives

$$K[7] = \frac{2^{5/7} 7^{3/4}}{14} \frac{\beta(1/7)\beta(2/7)}{\beta(1/14)}, \quad k_7^2 = \frac{8 - 3\sqrt{7}}{16}, \quad (103)$$

where  $\beta(t) = \Gamma(t)^2/\Gamma(2t)$ . We spell out the simplification because confusing the elliptic modulus with its square would give an incorrect constant.

Set  $g_1 = \Gamma(1/7)$ ,  $g_2 = \Gamma(2/7)$ ,  $g_4 = \Gamma(4/7)$ , and  $g_{1/2} = \Gamma(1/14)$ . Then

$$\frac{\beta(1/7)\beta(2/7)}{\beta(1/14)} = \frac{g_1^3 g_2}{g_4 g_{1/2}^2}.$$

The gamma duplication formula at  $1/14$  is

$$g_{1/2} g_4 = 2^{6/7} \sqrt{\pi} g_1.$$

It follows that the beta quotient is  $g_1 g_2 g_4 / (2^{12/7} \pi) = \mathcal{G}_7 / (2^{12/7} \pi)$ . Substitution into (103) gives

$$K[7] = \frac{7^{3/4} \mathcal{G}_7}{28\pi} = \frac{\mathcal{G}_7}{4\pi 7^{1/4}},$$

which is exactly (34). The classical table evaluation is an external input; the algebraic conversion here checks that its normalization matches the one used in the proof.

## B Two derivations from manuscript 56

### B.1 The recurrence-to-ODE identity

For completeness, write  $\vartheta = z d/dz$ . Then

$$\sum_{n \geq 0} n a_n z^n = \vartheta A, \quad \sum_{n \geq 0} n^2 a_n z^n = \vartheta^2 A.$$

The left side of (1), summed over  $n \geq 0$ , is  $A' + zA''$ . The first term on the right gives  $26z^2 A'' + 39zA' + 2A$ , and the shifted term gives  $27z^3 A'' + 54z^2 A' + 6zA$ . Their difference is exactly (17).

### B.2 A compact derivation of the coefficient algorithm

Assume

$$a_n = C 27^n n^{-3/2} \mathcal{A}_n, \quad \mathcal{A}_n \sim S_{27}(1/n).$$

Divide (1) by  $C 27^n n^{1/2}$  and put  $t = 1/n$ . The three terms become

$$27(1+t)^{1/2} S_{27}\left(\frac{t}{1+t}\right), \quad (26+13t+2t^2) S_{27}(t), \\ \left(1-t+\frac{2}{9}t^2\right) (1-t)^{-3/2} S_{27}\left(\frac{t}{1-t}\right),$$

which is  $\mathcal{E}_{27}[S_{27}] = 0$  in (74). The coefficient of  $t^2$  is  $-215/36 - 28c_1$ , giving  $c_1 = -215/1008$ ; successive coefficients proceed triangularly: after the automatic cancellations in degrees 0 and 1, each next coefficient determines one new  $c_j$  linearly. This proves that all  $c_j$  are rational and supplies a fast exact algorithm.

## C Possible OEIS formula updates

Both manuscripts propose an update; neither was submitted. The following is suggested mathematical content for an update after independent review. No edit to OEIS has been made as part of this work.

## C.1 Manuscript 58's proposal

Let  $G = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7)$ . Then

$$a(n) \sim \frac{\sqrt{3\pi}}{G} \frac{27^n}{n^{3/2}}, \quad \sum_{n \geq 0} \frac{a(n)}{27^n} = \frac{3G}{8\pi^2}.$$

More precisely, for every fixed truncation order there is an asymptotic expansion with a remainder of the next order. Its first corrections are

$$1 - \frac{215}{1008n} - \frac{1265}{290304n^2} + \frac{4683055}{877879296n^3} + O(n^{-4}).$$

The first three corrections were already calculated formally by O'Brien [3].

The separate Lucas-congruence statement is not affected by this report.

## C.2 Manuscript 56's proposal

A compact statement suitable for adaptation to the OEIS entry after independent review (its plain-text form is `data/56-exact-OEIS_update.txt`):

Let  $\mathcal{G}_7 = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7)$ . Then

$$a(n) \sim \frac{\sqrt{3\pi}}{\mathcal{G}_7} 27^n n^{-3/2}.$$

More precisely,

$$a(n) = \frac{\sqrt{3\pi}}{\mathcal{G}_7} 27^n n^{-3/2} \left( 1 - \frac{215}{1008n} - \frac{1265}{290304n^2} + \frac{4683055}{877879296n^3} + O(n^{-4}) \right).$$

For the generating function  $A(x) = \sum_{n \geq 0} a(n)x^n$ ,

$$A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}, \quad A(1/125) = \frac{5\mathcal{G}_7}{16\pi^2}.$$

At the dominant singularity,

$$A(x) = \frac{3\mathcal{G}_7}{8\pi^2} - \frac{2\pi\sqrt{3}}{\mathcal{G}_7} \sqrt{1-27x} + O(1-27x).$$

[Merge note, batch 7302: The first three corrections were calculated formally by O'Brien (2016 thesis, §8.4) [3]. Manuscript 56's proposal and its plain-text file do not say so; any submission must add this credit.]

## References

- [1] The OEIS Foundation Inc., *A279619: Expansion of the generating function of A002652 in powers of the generating function of A279618*. Entry by Lynette O’Brien, with later contributions; consulted October 1, 2026. The displayed entry revision was August 22, 2026. <https://oeis.org/A279619/internal>.
- [2] The OEIS Foundation Inc., *A183204*, recurrence, differential equation, generating-function relations, and asymptotic formula. Consulted October 1, 2026. <https://oeis.org/A183204/internal>.
- [3] L. A. O’Brien, *Modular forms and two new integer sequences at level 7*, Master of Science thesis, Massey University, 2016. Especially Theorem 6.1 and Section 8.4, pp. 63–67.  
<https://oeis.org/A279619/a279619.pdf>;  
<https://mro.massey.ac.nz/items/ad1bd663-2b56-4eed-ad1a-f130bb7fe757>.
- [4] M. D. Hirschhorn, *A sequence considered by Shaun Cooper*, New Zealand Journal of Mathematics **43** (2013). <https://web.maths.unsw.edu.au/~mikeh/webpapers/paper182.pdf>.
- [5] A. J. van der Poorten and K. S. Williams, *Values of the Dedekind eta function at quadratic irrationalities*, Canadian Journal of Mathematics **51** (1999), no. 1, 176–224.  
<https://doi.org/10.4153/CJM-1999-011-1>.
- [6] R. Hemmecke, P. Paule, and C.-S. Radu, *Computer-assisted construction of Ramanujan–Sato series for  $1$  over  $\pi$* , manuscript dated January 31, 2025. Equation (98), p. 32, is the level-seven identity used here.  
[https://www3.risc.jku.at/publications/download/risc\\_7134/.RamanujanSatoOneOverPi.pdf](https://www3.risc.jku.at/publications/download/risc_7134/.RamanujanSatoOneOverPi.pdf).
- [7] J. M. Borwein and I. J. Zucker, “Fast evaluation of the gamma function for small rational fractions using complete elliptic integrals of the first kind,” *IMA Journal of Numerical Analysis* **12** (1992), 519–526.  
<https://academic.oup.com/imajna/article-abstract/12/4/519/690281>.
- [8] I. J. Zucker, *Singular value theory*, exercise notes in Jonathan Borwein’s archived materials, Table 1, entry  $K[7]$ . The notes refer to Borwein and Zucker’s elliptic-integral evaluations; consulted October 1, 2026. <https://carmamaths.org/jon/Preprints/Books/EMA/Exercises/For%20others/K-beta.pdf>.
- [9] S. Cooper, *Sporadic sequences, modular forms and new series for  $1/\pi$* , Ramanujan Journal **29** (2012), 163–183.
- [10] D. A. Cox, *Primes of the Form  $x^2 + ny^2$* , 2nd ed., Wiley, 2013.
- [11] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., Springer, 1990.
- [12] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009. Chapter VI: singularity analysis and transfer of local expansions.
- [13] F. W. J. Olver et al., eds., *NIST Digital Library of Mathematical Functions*: Sections 4.13, 5.11, and 15.10 (Lambert’s  $W$  function, gamma asymptotics, the hypergeometric differential equation), cited by manuscript 58; Chapters 5, 23 and 27 (gamma function, modular functions, functions of number theory), listed by manuscript 56. Consulted October 1, 2026. <https://dlmf.nist.gov/>.
- [14] H. Cohen and W. Zudilin, *Continued fractions and irrationality measures for Chowla–Selberg gamma quotients*, arXiv:2510.00215, 2025, revised versions through 2026. Related work, not a theorem dependency here. <https://arxiv.org/abs/2510.00215>.

- [15] V. Reshetnikov, *ProveIt*, *Analysis/Transseries/README.md*, repository snapshot a5ad4ac2e2d4e642a6c015836a70492d40fcbc71.  
<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/Transseries>.
- [16] ProveIt, *Combinatorial Transseries and Their Inverses*,  
*Analysis/Transseries/docs/series-and-transseries/*  
*Combinatorial\_Transseries\_Inverses/*, Theorem t2:thm:balanced-inverse (explicit centered inverse coefficients). Cited in the merge; neither manuscript cites it.
- [17] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*,  
*Analysis/Transseries/docs/series-and-transseries/ Transseries\_And\_Inversion/*,  
Theorem p0:thm:staircase (staircase recovery and the separation condition). Cited in the merge.